Amortized Bayesian parameter recovery for dynamical systems with conditional flow matching

The `amortix` package: architecture, evaluation methodology, and benchmarks

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Abstract

We describe `amortix`, a package for amortized Bayesian estimation of the parameters of dynamical systems — ordinary and stochastic differential equations and discretized PDEs — from noisy observations at a finite set of time points. A conditional flow-matching model is trained once on simulated (parameter, data) pairs and thereafter produces posterior samples for any new observation set without likelihood evaluations or per-instance optimization; a single trained network handles any number and placement of observation points, the setting required for sequential experiment refinement and Bayesian experimental design. The package couples three components that we argue are only useful together: (i) a set-transformer encoder whose input representation is learnable rather than prescribed — a monotone mixture-of-scales reparametrization of the value axis, attention phases computed from physical observation times, and explicit set-cardinality features; (ii) a benchmark suite of fifteen systems spanning finance, neuroscience, epidemiology, ecology, pharmacokinetics, classical mechanics, and reaction–diffusion PDEs; and (iii) an evaluation harness in which simulation-based calibration is used only as a screen, and the instrument of record is comparison against exact or validated Markov chain Monte Carlo reference posteriors on the same observed points. Trained once per system in this design-amortized mode, the default network matches exact or validated reference posteriors to median Fréchet distances (in reference-normalized coordinates) of 0.01–0.07 on five of six reference-equipped systems — evaluated across design sizes with a fresh random design per test observation set, where it is comparable to or better than networks trained for each single design — while the single elevated entry, dense-design pharmacokinetics, is a measured effect: its posterior contracts to within a few multiples of the network's absolute error floor, which stays at the few-$\times 10^{-3}$ level of the prior range across systems whose posterior widths differ by 30×. Reaching those numbers required auditing the reference posteriors themselves — on the jump-diffusion, chains of the length we had been using disagreed with each other more than the model disagreed with them, and lengthening them moved the reported error from 0.19 to 0.05; 34 of 36 parameters across the nine systems without a validated reference pass simulation-based calibration, and residual errors decay steadily with the simulation budget. We compare against neural posterior estimation and flow-matching posterior estimation from the `sbi` package, both network families of BayesFlow 2, and a port of the Simformer, on three systems with exact or validated references. On a multiplicative process, all external configurations either inflate the volatility posterior by factors of 1.4–3.2 or become overconfident (widths 0.4–0.9) depending on the input coordinates they are given, while the learnable representation recovers the exact posterior from raw data. On an additive control process the same external tools improve markedly — the systematic bias disappears and the width inflation shrinks from 2.5–3.2 to 1.6–1.9 — which localizes the dominant failure to input conditioning while showing that a residual representation gap remains even in the benign case. We document this failure mechanism — the conditioning of the encoder input — in three distinct forms, together with learnable modules that repair each one.

1 Introduction

Recovering the parameters of a differential-equation model from noisy measurements of its solution is one of the basic inverse problems of applied science: a modeler writes down a dynamical system — an ODE, an SDE, or a PDE — whose right-hand side depends on unknown physical constants, observes a trajectory at a finite set of time points, and asks what the observations say about the constants. Like most inverse problems, this one is ill-posed in the practical sense: different parameter values can produce nearly indistinguishable observations, especially when the state is partially observed or the sampling is sparse, so a point estimate both hides the ambiguity and is unstable under noise. The Bayesian formulation resolves this by reporting the full posterior distribution over parameters given the data — ambiguity becomes posterior spread and correlation, quantities one can act on — at the price of two computational obstacles. First, for most stochastic and partially observed systems, the likelihood of the observations requires integrating over unobserved trajectory segments and is intractable or expensive. Second, modern workflows need posteriors *repeatedly*: calibration studies process thousands of synthetic observation sets, sequential experiments update the posterior after every new measurement, and Bayesian experimental design compares candidate observation schedules by the posteriors they would produce — so per-instance MCMC or optimization inside such loops is the dominant cost even when a likelihood is available.

Simulation-based inference (SBI) addresses both issues by learning a conditional generative model of the parameters given the data from simulated pairs — parameters drawn from the prior, observation sets produced by the simulator — thereby amortizing the cost of inference into a one-time training phase (Cranmer et al., 2020). Neural posterior estimation (NPE) and its sequential variants (Papamakarios and Murray, 2016; Greenberg et al., 2019) fit normalizing flows by maximum likelihood; flow-matching and diffusion variants (Wildberger et al., 2023; Gloeckler et al., 2024) replace the density estimator with transport-based generative models. Mature software exists: the `sbi` toolkit (Boelts et al., 2024), BayesFlow (Radev et al., 2020), and the `sbibm` benchmark collection (Lueckmann et al., 2021).

This report describes `amortix`, a compact PyTorch package (about 3,500 lines; one default model of 4.8×10^5 weights) for amortized Bayesian parameter recovery in dynamical systems. Three aspects of its design distinguish it from the toolkits above, and the report is organized around measuring their consequences.

The first concerns the interface between raw data and the network. Conditional density estimators are conventionally fitted to a fixed-length data vector after global standardization; the estimator sees whatever coordinates the user chose. We show in Sec. 8.3 that on multiplicative and heavy-tailed processes this convention fails in a specific, reproducible way across four different generative engines from three codebases, and that supplying hand-derived coordinates does not repair it uniformly: with identical inputs, different implementations end up over-dispersed, overconfident, or biased. In `amortix` the transformation from observations to network inputs is itself part of the trained model — a monotone mixture-of-scales reparametrization of the value axis, observation times entering attention as continuous phases, and explicit design-size features (Sec. 4); each component is motivated by a measured failure and validated by an ablation.

The second concerns observation designs. `amortix` trains a single network to approximate $p(m \mid S)$ coherently for every subset S of the measurement points of an experiment — arbitrary sizes and placements, including spatio-temporal sensor designs for PDEs (Sec. 8.5). This is the object consumed by sequential refinement and Bayesian experimental design, and supporting it changes the training procedure (Sec. 5): designs must be resampled fresh at every optimizer step, and the distribution of design sizes becomes a modeling choice with measurable consequences.

The third concerns evaluation. Simulation-based calibration (Talts et al., 2018) is a necessary-condition test whose statistical power at practical sample sizes is limited; we measure that limit (Sec. 6) and use SBC only as a screen. The instrument of record throughout is the comparison of posterior mean and width against an exact or validated-MCMC reference posterior on the same observed points, and the package ships this machinery — closed forms, exact-likelihood adaptive Metropolis samplers, chain validation — as part of its public API. The concern motivating this discipline is documented in the SBI literature: learned posteriors can be confidently wrong in ways that rank diagnostics do not detect (Hermans et al., 2022).

The remainder of the report presents the method (Sec. 3), the architecture (Sec. 4), training procedures (Sec. 5), the evaluation methodology (Sec. 6), the benchmark suite (Sec. 7), experimental results including comparisons with `sbi`, BayesFlow, and the Simformer on three systems with exact or validated references (Sec. 8), an analysis of the failure modes encountered during development (Sec. 9), and limitations (Sec. 10). Appendix A defines every benchmark system with its equations, priors, and observation model; Appendix C details the reference posteriors; Appendix D maps the terminology of this report to package options.

2 Problem formulation

Let a dynamical system depend on a parameter vector $m \in \mathbb{R}^d$ with a known prior $p(m)$, uniform on a box in all benchmarks below. The system produces a state trajectory $x(t)$ — through an ODE $\dot{x} = f(x; m)$, an SDE $dx = f(x; m)\, dt + \sigma(x; m)\, dW_t$, or a spatially discretized PDE — and the trajectory is observed through a noisy channel at K points:

$$y_i = h\big(x(t_i)\big) + \varepsilon_i, \qquad \mathcal{D} = \{(t_i, y_i)\}_{i=1}^{K}. \tag{1}$$

We call $\mathcal{D}$ the *observation set* of one experiment; it is the entire input to inference. Training and evaluation below involve many independently simulated observation sets, each paired with the parameter that generated it. The observation operator h may expose only part of the state: in the benchmarks below, only the price of an asset while its stochastic volatility is latent, only the membrane voltage of a neuron while its gating variables are latent, or only three spatial sensors of a PDE field. The target is the posterior $p(m \mid \mathcal{D}) \propto p(\mathcal{D} \mid m)\, p(m)$.

Three standing assumptions delimit the setting. The parameter is finite- and low-dimensional ($d = 2$–5 in every benchmark below) with a prior that factorizes across components and has tractable marginal CDFs (uniform boxes throughout; the construction extends to any such prior). The simulator can be run forward cheaply and repeatedly, but nothing else is required of it — no likelihood evaluations, no gradients. Observation times (and sensor indices) are known exactly; the measurement noise ε_i is independent across observations with a known per-system law.

Running example. The Ornstein–Uhlenbeck process is used throughout to make constructions concrete: $dX = -\theta X\, dt + \sigma\, dW$ with $m = (\theta, \sigma)$, prior box $[0.3, 3] \times [0.2, 1.5]$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$. An observation set is, for instance, $K = 74$ noisy values of X on a fixed two-rate grid, or — in the variable-design setting — any K values at arbitrary times. Its posterior has the features that recur across the suite: the diffusion coefficient σ is identified sharply by high-frequency increments; the reversion rate θ only weakly over a finite horizon; and the width of both marginals must shrink as observations accumulate. An estimator can therefore be wrong in the posterior mean, in the posterior width, or in the width-versus-K trend, and the evaluation methodology of Sec. 6 is built to detect all three.

Although the posterior itself is intractable, sampling from the *joint* distribution of parameters and data is easy: draw $m_j \sim p(m)$, simulate the system, apply the observation channel. This yields arbitrarily many training pairs $(m_j, \mathcal{D}_j)$, and a conditional generative model trained on them approximates the posterior for every observation set simultaneously — the cost of inference is moved into a one-time training phase, and inference for a new observation set is a forward pass. We impose one requirement beyond standard amortization: a single trained network must remain correct for any number K and any placement of observation times (and, for PDEs, sensor positions). Formally, for every subset S of the measurement points of one experiment the network must approximate $p(m \mid S)$, and the family of answers must be coherent — in particular the posterior must widen as points are removed. We refer to this requirement as *design amortization*; it is the object consumed by sequential refinement and by Bayesian experimental design, where candidate observation schedules are ranked by the posteriors they would produce.

3 Posterior sampling by conditional flow matching

The idea of the method is to represent the posterior not by a density formula but by a *transport*: a learned, data-conditioned deformation that carries a fixed reference distribution onto the posterior. Sampling then amounts to drawing from the reference and applying the deformation — there is no normalizing constant to compute and no invertibility constraint to build into the network, because the deformation is defined implicitly as the flow of a velocity field, and the field itself is fitted by ordinary least-squares regression on simulated pairs. This construction is conditional flow matching (Lipman et al., 2023).

Concretely, parameters are first reparametrized componentwise,

$$m \;\longmapsto\; \Phi^{-1}\big(F_{\text{prior}}(m)\big) \in \mathbb{R}^d, \tag{2}$$

where F_{prior} collects the marginal prior CDFs and Φ is the standard normal CDF. In the new coordinates the prior is exactly $\mathcal{N}(0, I)$ and box constraints disappear; usefully, the uninformative limit then costs nothing — when the data carry no information the posterior coincides with the reference and the transport is the identity. From here on we work exclusively in these coordinates and keep writing m for the parameter in them; the inverse of (2) is applied exactly once, at the end of sampling, to return draws on the original scale.

The transport is indexed by a *virtual time* $\tau \in [0, 1]$ — an algorithmic variable with no relation to the physical time t at which the dynamical system is observed; t appears only inside observation sets, τ

only in the transport. Subscripts in τ denote the position along the transport path: m_1 is a parameter drawn jointly with its observation set $\mathcal{D}$ (the endpoint), $m_0 \sim \mathcal{N}(0, I)$ is pure reference noise — not the transform of any parameter, but the raw material that the transport will have turned *into* a posterior draw by $\tau = 1$ — and between them lies the training interpolation $m_\tau = (1-\tau)\, m_0 + \tau\, m_1$. The velocity network is trained to predict the direction of this path by regression:

$$\mathcal{L}(\theta) = \mathbb{E}_{m,\mathcal{D},\; m_0\sim\mathcal{N}(0,I),\; \tau\sim U[0,1]} \big\| v_\theta(m_\tau,\, \tau,\, \mathcal{D}) - (m_1 - m_0) \big\|^2. \tag{3}$$

The minimizer of (3) is the marginal velocity field whose flow transports $\mathcal{N}(0, I)$ at $\tau = 0$ to $p(m_1 \mid \mathcal{D})$ at $\tau = 1$ (Lipman et al., 2023). Sampling draws $m_0 \sim \mathcal{N}(0, I)$, integrates $dm/d\tau = v_\theta(m, \tau, \mathcal{D})$ from $\tau = 0$ to 1, and maps the endpoint back through the inverse of (2); the numerical settings of the integration are implementation details, collected in Sec. 5. The same estimator family, with a different conditioning architecture, is available in `sbi` as FMPE (Wildberger et al., 2023); the comparison of Sec. 8.3 therefore isolates the effect of the input representation from the choice of generative engine.

No likelihood evaluations, simulator gradients, or per-instance optimization are required at any stage; the simulator is executed only forward, during training-set generation.

The training objective is consistent in the following sense.

Proposition 1 (consistency)**.** *Let $(m, \mathcal{D}) \sim p(m)\, p(\mathcal{D} \mid m)$ and let m_1 be given by* (2)*. If the loss* (3) *is minimized over all measurable velocity fields, then for $p(\mathcal{D})$-almost every $\mathcal{D}$ the flow of the minimizer transports $\mathcal{N}(0, I)$ at $t = 0$ to the conditional law $p(m_1 \mid \mathcal{D})$ at $t = 1$; equivalently, the sampling procedure above draws exactly from the posterior (Lipman et al., 2023; Wildberger et al., 2023).*

The proposition is an idealization: with a finite network, a finite simulation budget, and a numerical solver, no calibration guarantee survives, and the residual error becomes an empirical quantity. Section 6 builds the instruments that measure it.

4 Conditioning architecture

The velocity field $v_\theta(m, \tau, \mathcal{D})$ must be conditioned on an observation set of time-stamped measurements, and three properties of that set drive the architecture: the number of observations K varies from experiment to experiment (by two orders of magnitude in the variable-design setting); the observation times are arbitrary real numbers, in general not equally spaced, so positional schemes built for uniform sequences do not apply; and the times carry all ordering information, so the output must not depend on how the observations happen to be stored. These properties rule out the fixed-length-vector conditioners that are the default in NPE implementations, and they are naturally met by a transformer operating on one feature vector per observation — throughout this section, a *token.*

4.1 Overall scheme

The network evaluates $v_\theta(m, \tau, \mathcal{D})$, where m is the normalized parameter vector partway along its transport path — the interpolation m_τ at training time, the current ODE state at sampling time — and τ is the virtual time of the transport. The design follows the encoder–decoder pattern that is standard in conditional flow-matching and diffusion models: the data are encoded once, and a decoder combines that encoding with the object being generated,

$$v_\theta(m, \tau, \mathcal{D}) \;=\; \mathrm{Dec}\big(m,\; \tau,\; \mathrm{Enc}(\mathcal{D})\big), \tag{4}$$

with explicit types: Enc maps the observation set — the K time-stamped measurements of one experiment — to a memory of K vectors, $\mathrm{Enc}(\mathcal{D}) \in \mathbb{R}^{K\times d_{\mathrm{model}}}$, computed once per observation set, and $\mathrm{Dec} : \mathbb{R}^d \times [0, 1] \times \mathbb{R}^{K\times d_{\mathrm{model}}} \to \mathbb{R}^d$ combines the current transport point and the virtual time with that memory. Two differences from the image and text instances of this pattern dictate the concrete form (Fig. 1). First, the conditioning is not a fixed-size embedding but an observation set whose size K varies: Enc is therefore a transformer over per-observation tokens, its output is a *memory* of K encoded states rather than one pooled vector, and Dec reads the memory by cross-attention — the one standard mechanism indifferent to K. Second, the generated object is not an image but a short vector of d parameters, so the decoder is small: the d components of m are treated as d tokens, self-attention among them expresses posterior correlations (the ridge of the Fisher–KPP posterior, the drift–volatility coupling of GBM), and the virtual time τ enters every block through zero-initialized adaptive layer normalization

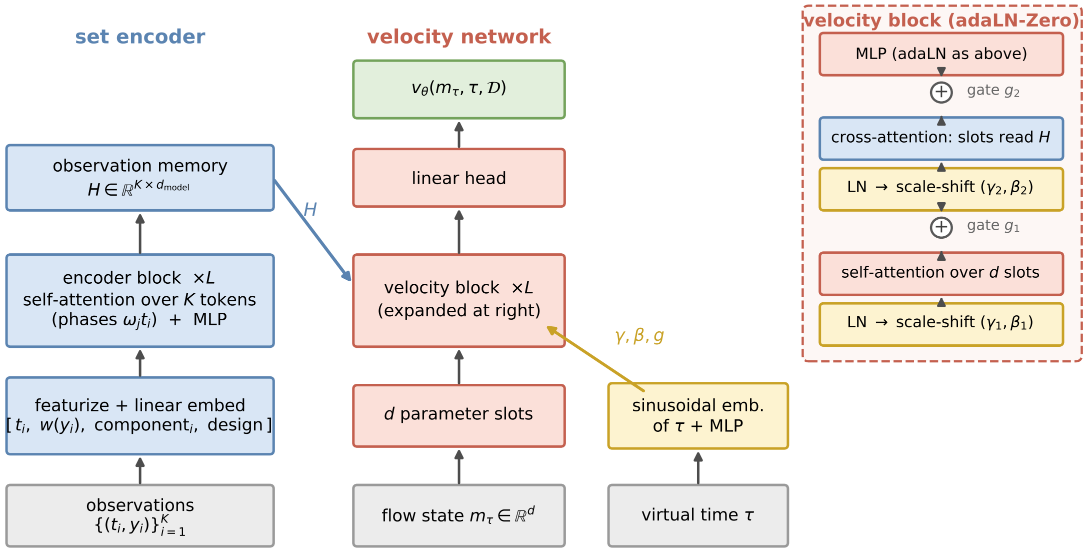

Figure 1: Model architecture; arrows denote data flow, bottom to top. Left: each observation is featurized and linearly embedded into one token; L transformer encoder blocks, whose attention phases are computed from the physical observation times (Sec. 4.2), produce a memory H of K encoded states. Middle: the d components of the flow state m_τ form d parameter-slot tokens processed by L velocity blocks and a linear head into the velocity v_θ. Right: one velocity block expanded — self-attention across parameter slots, cross-attention reading the observation memory H, and an MLP, each preceded by layer normalization with scale-shift (γ, β) and followed by a gated residual connection (g, marked +); the triples (γ, β, g) are produced from the virtual time τ by a zero-initialized map, so the untrained block is the identity (Peebles and Xie, 2023).

(adaLN-Zero), the conditioning mechanism of diffusion transformers (Peebles and Xie, 2023), so that the untrained network is an identity-like map.

On the encoder side, observation i is described by a few features — its time t_i, its transformed value $w(y_i)$ with w the learnable monotone map of Sec. 4.3, an index identifying the sensor or state component that produced the value (multivariate states contribute one token per observed component per time), and the design-size feature of Sec. 4.3 — and lifted to the model width by a learned linear map. The attention mechanism that lets these tokens interact is the subject of the next subsection; the choice of features is the subject of Sec. 4.3.

This scheme was not the first attempt but the survivor of a sequence of measured alternatives; the failed ones are retained in the package as reproducible controls and are compared under a common protocol in the ablation study of Appendix B. All sizes are configuration parameters. One default configuration (width 64, four heads, three blocks per side) is used unchanged for every experiment in this paper; at the training-set sizes we use, models of this scale proved sufficient — doubling width and depth changes no benchmark result (Sec. 8.4) — and nothing in the construction caps the scale.

4.2 Attention phases from observation times

We adapt rotary position embeddings — the standard positional scheme of modern transformers (Su et al., 2024) — to irregular time sampling.

First the classical construction. Let $x_1, \ldots, x_K \in \mathbb{R}^{d_{\text{model}}}$ be the token states entering an attention layer. Each head computes queries, keys, and values by learned linear maps,

$$q_i = W_Q x_i, \qquad k_i = W_K x_i, \qquad u_i = W_V x_i, \qquad W_Q, W_K, W_V \in \mathbb{R}^{d_h \times d_{\text{model}}}, \tag{5}$$

and token i aggregates the others with weights $a_{il} = \operatorname{softmax}_l\big(\hat{q}_i^\top \hat{k}_l / \sqrt{d_h}\big)$ applied to the u_l, where $\hat{q}, \hat{k}$ are position-modulated versions of q, k defined as follows. Split each query and key into $d_h/2$ two-dimensional blocks, $q_i = (q_i^{(1)}, \ldots, q_i^{(d_h/2)})$ with $q_i^{(j)} \in \mathbb{R}^2$, and rotate block j by an angle ϕ_i proportional to a per-token scalar:

$$\hat{q}_i^{(j)} = R(\omega_j \phi_i)\, q_i^{(j)}, \qquad \hat{k}_i^{(j)} = R(\omega_j \phi_i)\, k_i^{(j)}, \qquad R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}. \tag{6}$$

In the classical scheme the scalar is the token's integer position in the sequence, $\phi_i = n_i$, and the frequencies form a geometric ladder across blocks, $\omega_j = \omega_{\max}(\omega_{\min}/\omega_{\max})^{(j-1)/(d_h/2-1)}$, so that high-frequency blocks resolve neighboring positions and low-frequency blocks long ranges. Because rotations preserve inner products up to the relative angle,

$$\hat{q}_i^\top \hat{k}_l \;=\; \sum_{j=1}^{d_h/2} \big(q_i^{(j)}\big)^\top R\big(\omega_j(\phi_l - \phi_i)\big)\, k_l^{(j)}, \tag{7}$$

every attention logit depends on the positions only through differences $\phi_l - \phi_i$ — relative position awareness with no positional parameters.

For irregularly sampled data the integer position is meaningless, and the generalization is immediate: we set $\phi_i = t_i$, the physical time stamp of observation i, and choose the frequency range so that the periods $2\pi/\omega_j$ span from the finest observable time gap to several times the horizon. Nothing else changes; by (7), every logit now depends on the time stamps only through differences $t_l - t_i$. The substitution of physical time for position is the same one that time-aware attention models for irregular clinical and event series make in other positional schemes (Horn et al., 2020; Shukla and Marlin, 2021). Two consequences follow at once: the encoder is exactly permutation-invariant (relabeling tokens leaves the right-hand side of (7) unchanged; verified numerically to 10^{-8}), and variable designs require no positional bookkeeping. The phases enter only the encoder's self-attention; the cross-attention from parameter slots to the observation memory is unmodified attention, since parameter slots carry no time stamp and the memory states already encode temporal structure.

The adaptation is also motivated by a measured failure. An earlier configuration used integer-position rotations, trained with long token sequences subsampled in place (kept tokens retained their spread-out positions), and evaluated on contiguously packed short sequences. Every component was individually correct, yet the mismatch between the positional distributions seen at training and evaluation produced systematic posterior biases of $+1.1$ to $+3.3$ standard deviations (measured in Appendix B). Integer positions create an implicit train/test contract on token placement; physical-time phases eliminate the contract rather than manage it.

4.3 Input features

The remaining design choices concern what the tokens contain. Three of them matter and are described in turn: a learnable transform of the observed values, an optional increment-based token construction for directly observed diffusions, and an explicit design-size feature. Each occupies a few lines of code; each is load-bearing, as the ablations of Appendix B show.

Value transform. State variables of the benchmark systems live on very different scales, and some — prices, concentrations, populations — are multiplicative: the information sits in relative rather than absolute changes, and the coordinate in which such a variable is well-behaved is logarithmic. Rather than prescribing coordinates per problem, each value enters its token through a learned monotone map,

$$w(x) \;=\; \sum_{k=1}^{S} a_k \,\operatorname{sign}(x) \log\!\big(1 + |x|/s_k\big), \qquad a_k \ge 0. \tag{8}$$

Each basis function is linear for $|x| \ll s_k$ and logarithmic for $|x| \gg s_k$, so the family contains the identity, the signed logarithm, and every compression in between; positivity of the weights makes w monotone by construction; and — the operative property — every candidate compression is present from initialization, so training only reweights fixed nonlinearities, and the downstream network can exploit whichever coordinate helps from the first optimizer step. The scales s_k sit on a fixed geometric grid covering the dynamic range of the data ($S = 8$ octaves by default, one setting for all fifteen systems, no per-problem adjustment).

This structure has to earn its place against simpler-looking learnable alternatives, and we tested them on GBM, where the correct coordinate (log-price) is known and the cost of missing it is a measured $+0.48$ sd volatility bias. A generic two-layer MLP on the raw values, and a random-Fourier-feature map, both left the bias in place at matched budget: a free-form map must discover the shape and the scale of the compression simultaneously, from gradients arriving through the whole downstream network, and in practice it does not. A classical one-parameter statistical family — the Yeo–Johnson power transforms (Yeo and Johnson, 2000), in essence $x \mapsto ((1 + |x|)^\lambda - 1)/\lambda$ applied with sign symmetry, which passes through the identity at $\lambda = 1$ and tends to the logarithm as $\lambda \to 0$ — also failed, in an instructive way: trained end-to-end, λ never left its initialization, because a single global exponent must traverse a gradient plateau before the downstream network has learned anything that would reward the move. The mixture (8) removes the bias reliably, and on GBM the trained weights concentrate on the small-scale components: the model learns the log-price coordinate.

Increment features for directly observed diffusions. For scalar SDEs whose state is observed directly — no latent components — the package offers a second token construction: after sorting by time, consecutive observations contribute

$$\big[\, t_{(i)},\ w(y_{(i)}),\ \Delta w_i,\ (\Delta w_i)^2,\ w_t(\Delta t_{(i)}) \,\big], \qquad \Delta w_i = w(y_{(i+1)}) - w(y_{(i)}).$$

The motivation is elementary: for a Markov observed process the likelihood factorizes over consecutive pairs, transition densities of diffusions depend on increments and gaps, and squared increments carry the quadratic-variation information that identifies diffusion coefficients. This is an inductive bias of the same character as convolutions for images — useful where its assumption holds, not a requirement of the method: the point-token path of Sec. 4.1 is fully general, and the ablations of Appendix B quantify what the increment tokens buy at fixed budget.

The rule for choosing between the constructions — increment tokens exactly when the observed process is Markov — was tested in both directions: an increment-token model class transfers across Markov systems without adjustment (GBM to OU), and on the Heston model, whose observed price is not Markov because volatility is latent, increment and point tokens perform identically across all 25 evaluation cases, as the factorization argument predicts. The package selects the construction from a per-problem flag.

Two further normalization details matter for processes with jumps, where squared increments within a single observation set span several orders of magnitude and a global feature normalization becomes unstable — concretely, on the Merton benchmark the running normalization scale is set by rare jump outliers and the resulting representation drift produces a $+1.12$ sd volatility bias. First, increment features are logarithmically compressed before normalization; this is a near-identity on systems without jumps (since $\log(1 + x) \approx x$ for small x) and by itself turns the Merton failure into a calibrated posterior across all 25 evaluation cases (Sec. 9). Second, the normalization is made adaptive to the observation set: robust summary statistics s of the observation set, pooled over non-padding positions, are mapped by a zero-initialized MLP to per-channel scale and shift corrections,

$$h_c \ \mapsto\ \big(1 + \gamma_c(s)\big)\, h_c + \beta_c(s) \tag{9}$$

— a feature-wise affine modulation, FiLM in the terminology of Perez et al. (2018). Its measured contribution is specific and modest: it removes a residual $+0.27$ sd bias that survives on the densest Merton designs (the effective jump-detection threshold shifts with the unknown volatility), and it does so as well as a hand-crafted per-observation-set rescaling, within error bars ($+0.13 \pm 0.06$ vs $+0.09 \pm 0.05$ sd; Sec. 9). Three implementation conditions are necessary and were each found through a failure: zero initialization (training starts from the working baseline), modulation applied *after* the running normalization (the reverse ordering re-creates the drift it corrects), and pooled statistics that exclude padding positions.

Design size. Posterior width is governed by the number of observations — in regular models the error scales as $K^{-1/2}$ — but K is a global property of the set, and softmax attention, which computes normalized weighted means, is nearly invariant to it. Softmax attention computes normalized weighted means, which are nearly invariant to the number of tokens — yet posterior width is governed by K. The deficit is measurable by linear probes from the pooled encoder output: all architecture variants of Table 14 recover posterior-mean statistics ($R^2 = 0.83$–0.94), while $R^2(\log K)$ ranges from 0.985 to 0.498 and ranks the variants exactly in the order of their width errors. The repair is to make the count available, and the package default is the simplest form: $\log K$ enters every token as one of the input features. An alternative that requires no explicit feature — appending a few learned auxiliary tokens to the set, in the spirit of register tokens (Darcet et al., 2024), and letting the model accumulate the count in them — achieves the same effect in our ablations; the number of such tokens is a hyperparameter with no measurable influence beyond one. With the count available, the width error on the hardest case ($K = 100$) drops from 1.95 to 1.49 with bias unchanged; the residual is closed by the training procedure of the next section.

5 Training procedures

Common settings. Sampling integrates the velocity field with a fixed-step midpoint scheme, 20 steps by default; the step count was fixed once by a convergence check (fourth-order Runge–Kutta with three times as many steps changes no benchmark statistic), and evaluation uses an exponential moving average of the training weights. All models are trained with Adam at a base learning rate of 3×10^{-4} under a cosine schedule with linear warmup, batch size 64, and gradient clipping; evaluation uses an exponential moving average of the weights with decay 0.999. At every optimizer step the virtual times τ are drawn afresh (stratified over $[0, 1]$) together with fresh reference noise m_0, so no pairing between observation sets and virtual times is ever repeated. For fixed-design problems the training set consists of n simulated observation sets and training runs for a fixed number of epochs over it; the gallery of Sec. 8.2 uses $n = 40{,}000$ simulations and 35 epochs.

Design amortization. Training data are simulated once on a fine grid; observation designs are then resampled *from the stored fine trajectories at every optimizer step.* The procedure has three components, each fixed by a controlled comparison on three systems (GBM, Ornstein–Uhlenbeck, FitzHugh–Nagumo):

1. *Fresh designs at every step.* Re-drawing each trajectory's observation subset at each step makes design memorization impossible and lets one simulation serve arbitrarily many designs. The alternative — one fixed design per trajectory — closes the dense-design error at long budgets but overfits the design set, producing overconfidence at sparse K (width ratio 0.86 against the exact reference).
2. *The design-size law.* K is drawn 50% log-uniform on $[K_{\min}, K_{\max}]$ and 50% uniform on the dense half of the range. Log-uniform sampling alone places $\approx 6\%$ of the mass at $K \geq 100$ and starves the dense regime, which appears as inflated widths precisely there; the mixture alone halves the dense-design excess (GBM $1.27 \to 1.14$, OU $1.34 \to 1.14$; the FitzHugh–Nagumo dense-K SBC failure at $p < 10^{-3}$ disappears, $p = 0.54$).
3. *Budget scaling.* When higher accuracy is required, simulations and optimizer steps are scaled together at constant epochs; Sec. 8.4 shows the residuals then follow power laws in this single budget variable, and Sec. 8.4 measures why the scaling must be joint: passes over reused observation vectors eventually destroy accuracy, while fresh simulations do not.

With this procedure, the GBM posterior width matches the exact per-design posterior to 5–7% uniformly over $K = 5, 9, 20, 100$ (Fig. 7).

6 Evaluation methodology

Two instruments with different roles are used throughout, plus a diagnostic probe.

6.1 Simulation-based calibration as a screen

SBC (Talts et al., 2018) tests a necessary condition: for $m^* \sim p(m)$ and $\mathcal{D} \sim p(\mathcal{D} \mid m^*)$, the rank of each component of m^* among posterior samples must be uniform. We run SBC per parameter (uniformity test on 500 simulated observation sets with 200 posterior draws each unless stated; $p > 0.05$ passes) and report coverage of central 50% and 90% intervals.

The power of the test at this size is limited, and because our conclusions depend on knowing that limit, we measured it: on cases where an exact reference is available, SBC at 300–500 observation sets fails to reject posterior width ratios up to ≈ 1.3, and its per-parameter p-values are unstable between retrainings when residual biases are in the range 0.1–0.2 posterior standard deviations. SBC also cannot penalize a posterior that ignores the data and returns the prior; the prior-mean and ridge controls of Sec. 7 exist for that reason. SBC is therefore used as a cheap screen, never as a verdict.

6.2 Reference posteriors as the record

Wherever a tractable likelihood exists, we compare distributions on the same observed points. The headline statistic is a single number per observation set, the squared Fréchet distance between Gaussian fits of the two sample sets in reference-normalized coordinates (each parameter divided by the reference posterior standard deviation),

$$d_F^2 \;=\; \|\mu_a - \mu_r\|^2 + \mathrm{tr}\big(\Sigma_a + \Sigma_r - 2\,(\Sigma_r^{1/2}\Sigma_a\Sigma_r^{1/2})^{1/2}\big) \tag{10}$$

— the construction used as the FID score in the image-generation literature, and we adopt that name.

Every reference carries its own floor. A reference posterior is itself a finite sample, and if it is produced by MCMC it is a finite sample from a chain that may not have converged. We therefore compute, for every system and at the sample size actually reported, the same statistic between *two independent reference draws* — a second exact draw, or a second chain from a different seed. That number is the floor below which nothing can be measured, and it is quoted alongside the model's value throughout.

The check is not a formality: it caught a broken reference in this suite. The Merton jump-diffusion battery, built with 2,000-draw adaptive-Metropolis chains like every other MCMC reference here, has a two-chain floor of 0.259 — larger than the 0.186 we were reporting as the model's error, so that entry measured the sampler rather than the network. Lengthening the chains repairs it: at 20,000 draws the floor falls to 0.023 against a model value of 0.031, and at 10^5 draws to 0.005 against 0.021. The corrected references used in Sec. 8 are 2×10^5-draw chains thinned to 4,000 draws, whose two chains agree to 0.016 posterior standard deviations in the median and 0.060 at worst. No other system in the suite failed the check; the systems that cannot pass it at any chain length we could afford are reported under the calibration screen instead (Sec. 8.5).

Resolution of the estimator. Both sample sets are finite, so (10) is positive even for a perfect posterior, and the size of that offset sets what the metric can resolve. We measured it directly: drawing *two independent sample sets from the same exact GBM posterior* and scoring one against the other gives a median FID of 0.0157 at 500 samples, 0.0038 at the 2,000 used throughout this report, 0.0009 at 8,000, and 0.0002 at 32,000 — a clean $1/n$, as the estimation error of the fitted means and covariances predicts. Two consequences. Differences below ≈0.005 at $n = 2{,}000$ are not resolved, so values in that range are reported as "at the resolution limit" rather than compared. And the best values in this report sit above it rather than at it: the design-amortized GBM model reads 0.0107 at $n = 2{,}000$, 0.0100 at 8,000, and 0.0072 at 32,000 on the same battery, so after subtracting the estimator offset its excess over the reference is ≈0.007 at every sample size — a real error of the model, not an artifact of the ruler.

Does the choice of discrepancy matter? Equation (10) is the squared 2-Wasserstein distance between Gaussian fits, and the division by the reference standard deviation is ours, not part of that distance: the summands carry different physical units — a rate in h^{-1}, a volume in litres — so some normalization is required, and the one we chose contracts with the posterior itself. Table 1 therefore evaluates five discrepancies on the same sample sets: the Gaussian W_2^2 under two different normalizations, two distribution-free alternatives, and the classifier two-sample accuracy used as the primary metric of the `sbibm` benchmark (Lueckmann et al., 2021). Three readings. The distribution-free measures agree with the Gaussian one on every system, so the Gaussian summary is not what limits the comparison — the normalization is: the same model reads 0.009 on GBM and 0.58 on dense pharmacokinetics in posterior-sd units, but 0.0001 against 0.0003 in prior-range units. C2ST is the most readable of the five, being bounded and calibrated at 0.5: five of the six systems sit at 0.52–0.55, i.e. a trained classifier can barely separate our posterior samples from exact ones, with dense pharmacokinetics the single exception at 0.77. And the null column earns its place — it is what exposed the Merton reference above.

Why the image-style construction is not enough. In image generation, FID needs no likelihood because reference samples come straight from the dataset. The analogue here is a *joint-space* FID, which is likelihood-free too: draw $(m_i, \mathcal{D}_i)$ from the prior and the simulator, draw $m_i' \sim q(m \mid \mathcal{D}_i)$ from the trained model, and compare the two sets of pairs $\{(m_i, \mathcal{D}_i)\}$ and $\{(m_i', \mathcal{D}_i)\}$. We measured this construction on the GBM comparison of Sec. 8.3, where the exact per-design FID is known for every arm, and it fails as an instrument of record in both ways that matter: it compresses a 200× range of per-design error into 25× (this package 0.04 → 0.002; the worst external arm 8.5 → 0.03), and it misranks — NPE on raw prices (per-design FID 4.8) scores *better* in joint FID than NPE on hand-derived log-returns (per-design 1.5; joint 0.028 vs 0.045). The mechanism is averaging: Gaussian moments of the pooled joint blend per-observation-set errors, so width inflation dilutes and $\mathcal{D}$-correlated bias inflates, in different proportions per arm. A joint two-sample statistic is therefore a necessary-condition screen of the same tier as SBC, and comparisons on fixed observed points remain the record (per-arm joint values in `results/`). FID here is zero for a perfect match, dimensionless, and combines the three ways a posterior can be wrong: location error, width error, and correlation (orientation) error. For scale, a bias of 0.1 reference-sd in each of d dimensions gives FID $0.01\,d$, and a posterior 30% too wide in one dimension gives FID ≈ 0.09.

system	W_2^2 Gaussian / post. sd	/ prior range	distribution-free energy	sliced W_2^2	C2ST
geometric Brownian motion	0.009	0.0001	0.004	0.005	0.520
linear-Gaussian	0.043	0.0001	0.011	0.011	0.548
Ornstein–Uhlenbeck	0.050	0.0008	0.027	0.028	0.551
Fisher–KPP	0.071	0.0000	0.023	0.031	0.546
Merton jump-diffusion	0.035	0.0008	0.008	0.007	0.522
pharmacokinetics	0.580	0.0003	0.191	0.182	0.769
null: the same statistic between two independent reference draws					
geometric Brownian motion	0.002	0.0001	0.001	0.002	0.500
linear-Gaussian	0.004	0.0000	−0.001	0.002	0.498
Ornstein–Uhlenbeck	0.013	0.0003	0.003	0.009	0.507
Fisher–KPP	0.006	0.0000	0.004	0.008	0.510
Merton jump-diffusion	0.007	0.0003	0.000	0.002	0.497
pharmacokinetics	0.028	0.0000	0.008	0.014	0.527

Table 1: Five discrepancies between the same trained model and the same references, on all six reference systems, medians over 16–32 test observation sets with 4,000 samples each, and (lower block) the floor each statistic carries at that sample size. Energy and sliced W_2^2 are computed in posterior-sd units like the first column. The measures rank the systems identically; what separates them is the normalization, not the Gaussian assumption. Merton is scored against the converged reference of Sec. 6.2 (its 4,000-draw chains fail the two-chain check, which is why the floor column matters).

Where the *direction* of a failure matters — too wide is merely wasteful, overconfident is silently dangerous — we additionally quote its decomposition per parameter: the signed posterior-mean error in reference-sd units (*bias*) and the ratio of standard deviations (*width*; 1 exact, above one conservative, below one overconfident). References are of three kinds (full details in Appendix C):

- *Closed form.* The linear-Gaussian benchmark (exact Gaussian posterior) and geometric Brownian motion, whose log-returns give a conjugate normal–inverse-χ^2 posterior importance-corrected to the box prior.
- *Exact-likelihood MCMC.* An adaptive random-walk Metropolis sampler (after Haario et al., 2001): proposal scale adapted to a 0.25 acceptance target during burn-in, proposal shape replaced by the empirical covariance Cholesky factor rescaled to $2.38/\sqrt{d}$, adaptation frozen before the retained phase, out-of-box proposals rejected (which samples the box-truncated target exactly). Likelihoods are exact per-gap transition densities: Gaussian Euler–Maruyama chain transitions for OU (matching the simulator that generated the data, plus the stationary density of the informative initial state), a Poisson-mixture transition density for the Merton jump-diffusion, log-normal residuals around the Bateman curve for pharmacokinetics, and a Gaussian observation model around the deterministic PDE solve for Fisher–KPP — in the last case one likelihood evaluation is one PDE solve.
- *Chain validation.* A reference is used only after validation. Every MCMC reference in this report is run at least twice from well-separated starting points, and the between-chain discrepancy of posterior means (in posterior-sd units) is reported with the result it certifies. Where a closed form and a chain both exist, the chains reproduce it to a worst mean discrepancy of 0.099 sd and width ratios 0.91–1.03 over 24 comparisons.

Twice during development the screen and the record contradicted each other; both times the contradiction exposed an implementation error (a padding mask leaking into set-level statistics; the positional contract of Sec. 4.2). We consider the redundancy a feature of the methodology rather than an inefficiency.

6.3 Encoder probing

To separate encoder failures from flow failures, we fit ridge regressions from the pooled encoder output to known sufficient statistics of the problem. High probe R^2 with poor posterior accuracy localizes the defect downstream of extraction. In the Merton analysis of Sec. 9, the failing configuration extracted robust volatility statistics *better* than the healthy one ($R^2 = 0.967$ on truncated realized variance), redirecting the repair from the encoder's capacity to the stability of its input representation.

system	type	d	obs.	design	reference posterior
linear-Gaussian	linear map	4	full	fixed	exact Gaussian
Ornstein–Uhlenbeck	SDE	2	full	fixed	exact-likelihood MCMC
geometric Brownian motion	SDE	2	full	fixed	conjugate closed form
Cox–Ingersoll–Ross	SDE	3	full	fixed	— (Euler pseudo-MLE)
double well	SDE	3	full	fixed	—
stochastic Lotka–Volterra	2-d SDE	4	full	fixed	—
FitzHugh–Nagumo	2-d SDE	4	v only	fixed	—
SEIR	4-d SDE	5	I, D	fixed	—
polynomial-drift SDE	SDE	5	full	fixed	—
Heston	2-d SDE	5	price only	variable	—
Merton jump-diffusion	jump SDE	5	full	variable	Poisson-mixture MCMC
Hénon–Heiles	Hamiltonian ODE	3	q_1 only	variable	—
Hodgkin–Huxley	4-d ODE	4	V only	variable	—
pharmacokinetics (Bateman)	ODE (closed form)	3	concentration	variable	log-normal MCMC
Fisher–KPP	PDE (64-point grid)	2	3 sensors	variable	PDE-likelihood MCMC

Table 2: The benchmark suite. "obs." lists the observed coordinate when the state is only partially visible; "design" distinguishes the fixed-grid gallery from variable-design systems, where every observation set carries its own set of (time, sensor) observation points. Dashes in the last column indicate systems where no tractable reference exists — these are evaluated by SBC and controls only, which is precisely the regime that motivates carrying exact-reference systems in the same suite.

7 Benchmark suite

The suite contains fifteen systems (Table 2), defined below with their equations, the meaning and prior range of every parameter, and the observation model. Variable-design training — observation subsets resampled at every optimizer step — is the package default and the recommended mode: beyond enabling inference at any design, the resampling acts as data augmentation over designs. Nine systems are nevertheless benchmarked on conventional fixed grids, for two practical reasons: their closed-form or cheap references live on those grids, and the external tools of Sec. 8.3 consume fixed-length inputs.

All SDEs are integrated by Euler–Maruyama on the stated fine grid; ODEs by RK4 (Hodgkin–Huxley by a semi-implicit scheme for the gating variables); the PDE by finite differences. Systems marked *random designs* are trained and evaluated in the design-amortized mode of Sec. 2: the observation times (and, for the PDE, sensor indices) of every observation set are drawn at random, and one trained network serves all of them; the remaining systems use one conventional fixed grid. The *classical estimator* listed for each system is the standard non-Bayesian method a practitioner would apply; all of them return point estimates, not distributions, and they serve as an accuracy floor for the posterior mean. Three recur and are defined here once. *Kramers–Moyal regression* estimates the drift and squared diffusion from the conditional first and second moments of the observed increments ($f(x) \approx \mathbb{E}[\Delta X \mid X = x]/\Delta t$, $\sigma^2(x) \approx \mathbb{E}[\Delta X^2 \mid X = x]/\Delta t$) and fits the parametric form to these estimates by least squares. *Multi-start nonlinear least squares* fits the deterministic solve of the system to the observed trajectory by minimizing squared error over parameters, restarted from many random initial points to escape local minima. *Exact maximum likelihood* is available where the transition density is known in closed form. Fixed-design observations are token sets over the union of the stated channels; "fast/slow" channels observe every step over a short window and every j-th step over the horizon respectively. Priors are uniform on the stated boxes. Classical estimators (`sota()`) are listed per system.

Linear-Gaussian. $y = Am + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.5^2 I_6)$, with a fixed correlated design matrix $A \in \mathbb{R}^{6\times 4}$; $m \in [-3, 3]^4$. Exact Gaussian posterior; the classical estimator is the posterior mean itself. This system checks the flow engine in isolation from the encoder.

Ornstein–Uhlenbeck. $dX = -\theta X\, dt + \sigma\, dW$, stationary start $X_0 \sim \mathcal{N}(0, \sigma^2/2\theta)$; $dt = 0.02$, 500 steps; channels: fast (1×50), slow (20×24). Prior $\theta \in [0.3, 3]$, $\sigma \in [0.2, 1.5]$. Classical: exact MLE from per-gap Gaussian transitions.

Geometric Brownian motion. $dS = \mu S\, dt + \sigma S\, dW$, $S_0 = 1$; $dt = 0.01$, 500 steps (five "years"); channels: fast (1×60), slow (10×40). Prior $\mu \in [-0.2, 0.4]$, $\sigma \in [0.1, 0.6]$. Classical: exact MLE on log-returns. Reference: conjugate posterior (Appendix C).

Cox–Ingersoll–Ross. $dX = a(b - X)\,dt + \sigma\sqrt{X}\,dW$; $dt = 0.02$, 500 steps; channels as OU. Prior $a \in [0.3, 3]$, $b \in [0.3, 1.5]$, $\sigma \in [0.1, 0.5]$. Classical: Euler pseudo-likelihood MLE (Cox et al., 1985).

Double well. $dX = (\theta_1 X - \theta_2 X^3)\,dt + \sigma\,dW$; $dt = 0.01$, 1000 steps, noise strong enough for barrier crossings; channels: fast (1×80), slow (25×36). Prior $\theta_{1,2} \in [0.5, 3]$, $\sigma \in [0.4, 1.2]$. Classical: Kramers–Moyal regression (defined above).

Stochastic Lotka–Volterra. Prey/predator pair with multiplicative noise, $dx_1 = (\alpha x_1 - \beta x_1 x_2)\,dt + 0.1 x_1\,dW_1$, $dx_2 = (\delta x_1 x_2 - \gamma x_2)\,dt + 0.1 x_2\,dW_2$; both components observed; $dt = 0.01$, 600 steps; channels: fast (1×50), slow (15×40). Prior $\alpha, \gamma \in [0.8, 1.5]$, $\beta, \delta \in [0.4, 1.2]$. Classical: multi-start nonlinear least squares (defined above).

FitzHugh–Nagumo. $\dot{v} = v - v^3/3 - w + I + \text{noise}$, $\dot{w} = \epsilon(v + a - bw)$; only v observed at 25 evenly spaced times over horizon 40 ($dt = 0.05$), observation noise 0.05. Prior $a, b \in [0.5, 0.9]$, $\epsilon \in [0.05, 0.3]$, $I \in [0, 0.5]$. Classical: multi-start nonlinear least squares (defined above).

SEIR with mortality. Five-compartment stochastic epidemic (two-phase contact rate $\beta_1 \to \beta_2$, incubation α, recovery γ_r, mortality γ_d); infected and deceased compartments observed at 20 days over a 60-day horizon ($dt = 0.1$), observation noise 0.004. Prior $\beta_1 \in [0.2, 0.6]$, $\beta_2 \in [0.05, 0.3]$, $\alpha \in [0.2, 0.5]$, $\gamma_r \in [0.05, 0.2]$, $\gamma_d \in [0.005, 0.05]$. Classical: nonlinear least squares on (I, D).

Polynomial-drift SDE. $dX = (c_0 + c_1 X + c_2 X^2 + c_3 X^3)\,dt + \sigma\,dW$ — drift identification on a dense polynomial basis, the setting of SINDy (Brunton et al., 2016) without its sparsity prior (a sparse-coefficient variant is discussed in Sec. 10); $dt = 0.01$, 1000 steps; channels: fast (1×80), slow (20×45). Prior $c_0, c_2 \in [-0.5, 0.5]$, $c_1 \in [-2, 0]$, $c_3 \in [-1, -0.1]$, $\sigma \in [0.3, 0.8]$. Classical: least squares of the Kramers–Moyal drift estimate on the polynomial basis — the dense-basis version of SINDy regression.

Heston (random designs). $dS = \mu S\,dt + \sqrt{v}\,S\,dW_1$, $dv = \kappa(\bar{v} - v)\,dt + \xi\sqrt{v}\,dW_2$, $\mathrm{corr}(dW_1, dW_2) = \rho$ (Heston, 1993); price observed, volatility latent; $dt = 0.01$, 500 steps, designs of $K \in [2, 128]$ random times. Prior $\mu \in [-0.1, 0.3]$, $\kappa \in [0.5, 5]$, $\bar{v} \in [0.02, 0.3]$, $\xi \in [0.1, 1]$, $\rho \in [-0.9, 0]$.

Merton jump-diffusion (random designs). $dS/S = \mu\,dt + \sigma\,dW + d\big(\sum_i (e^{J_i} - 1)\big)$ with Poisson jump rate λ and $J_i \sim \mathcal{N}(\mu_J, s_J^2)$ (Merton, 1976); $dt = 0.01$, 500 steps, $K \in [2, 128]$. Prior $\mu \in [-0.1, 0.3]$, $\sigma \in [0.1, 0.5]$, $\lambda \in [0.2, 3]$, $\mu_J \in [-0.3, 0.3]$, $s_J \in [0.05, 0.4]$. Reference: Poisson-mixture transition density (Appendix C).

Hénon–Heiles (random designs). Two-dimensional Hamiltonian oscillator with the coupling potential used in Lubich et al. (2015), frequencies (ω_1, ω_2) and coupling λ; q_1 observed at random times with noise; $dt = 0.05$, 800 steps, $K \in [4, 64]$. Prior $\omega_{1,2} \in [0.7, 1.3]$, $\lambda \in [0.02, 0.22]$.

Hodgkin–Huxley (random designs). Standard four-dimensional membrane model (Hodgkin and Huxley, 1952); voltage observed at random times; parameters $(g_{\mathrm{Na}}, g_K, g_L, I)$ with prior $[60, 180] \times [10, 50] \times [0.05, 0.5] \times [4, 12]$; 3000 integration steps, $K \in [4, 96]$.

Pharmacokinetics (random designs). The standard one-compartment oral-dosing model of pharmacokinetics (e.g. Gabrielsson and Weiner, 2000): absorption at rate k_a, elimination at rate k_e, distribution volume V, giving the Bateman concentration curve, concentration $c(t) = \frac{D_0 k_a}{V(k_a - k_e)}\big(e^{-k_e t} - e^{-k_a t}\big)$ with dose $D_0 = 500$; log-normal assay noise (10%); horizon 24 h, $K \in [3, 64]$ irregular draws. Prior $k_a \in [0.5, 4]$, $k_e \in [0.05, 0.4]$, $V \in [20, 100]$. Reference: log-normal likelihood MCMC.

Fisher–KPP (random designs). $\partial_t \theta = D\,\partial_{xx}\theta + r\,\theta(1 - \theta)$ on $[0, 1]$ with reflecting boundaries, 64-point finite-difference grid, Gaussian initial bump; three fixed sensors at $x = 16/63, 32/63, 48/63$ observed at random (time, sensor) pairs with noise 0.02; $dt = 0.01$, 200 steps, $K \in [4, 96]$. Prior $D \in [0.001, 0.01]$, $r \in [2, 10]$. Reference: Gaussian observation model around the deterministic solve; one likelihood evaluation is one PDE solve.

Controls. Two model-free controls accompany every result. The *prior-mean predictor* ignores the data entirely (exactly 25% mean absolute error under a uniform prior). The *ridge control* is a point predictor of the parameters: a degree-2 ridge regression on generic summary statistics of the token set — per token feature, the mean, standard deviation, mean square, mean absolute value, and the 0.1/0.5/0.9 quantiles, computed over however many observations are present. It recovers no distribution and is not meant to; its role is to detect a specific silent failure that the calibration screen cannot see: a posterior that ignores the data and returns the prior passes SBC perfectly, and only a point-accuracy control unmasks it.

8 Experimental results

This section is organized as an experimental study: the protocol — references, baselines, model sizes, budgets — is fixed first; the main results (posterior overlays and the global metric on every system) follow; then the comparison with existing tools, the scaling behavior in budget, model size and data reuse, and the evaluation of design amortization.

8.1 Experimental protocol

References and baselines. Every accuracy number in this section is the FID of Sec. 6.2 against a reference posterior computed on the same observed points, reported as the median over the system's test battery (the distribution over observation sets is right-skewed; means are quoted where the tail is the finding, and for scale, a 0.1 sd bias in each of d dimensions gives FID $0.01\,d$). The reference samplers and their cost per observation set for 2,000 samples (laptop CPU; validation, where used, doubles it): the exact conjugate posterior for GBM (<1 ms), exact-likelihood MCMC for GBM and OU (42–124 ms), mixture-likelihood MCMC for Merton and pharmacokinetics (0.03–0.1 s), and PDE-likelihood MCMC for Fisher–KPP (271 s — ≈12,000 solves, the number of likelihood evaluations any MCMC spends per observation set whatever the forward model costs). The external amortized tools appear in the same tables as our models, at the shared 20,000-simulation budget of Sec. 8.3. Simulation-based calibration is reported as the number of parameters passing the uniformity screen at $p > 0.05$ over 500 simulated observation sets (*SBC pass*); with 32 such tests on the fixed-grid systems, one or two rejections are expected by chance even from an exact posterior.

Models and training. For each reference system we train the same architecture at three sizes — tiny (0.1M parameters), the default small (0.5M), and big (2.3M) — plus two miniature sizes (nano, 26k; pico, 7k) used to locate the capacity floor. All runs follow the procedure of Sec. 5, with one coupling that must be respected: at width 128 the base learning rate is halved, otherwise optimizer instability masquerades as a capacity effect (Sec. 8.4). Simulation budgets range from 5,000 to 480,000 per system. The full size × budget grid, with the external tools in the same table, opens the results (Table 3); the summary tables then use each system's best trained model. Observation designs are random at training and at evaluation wherever the system defines a design distribution; fixed designs appear only where the interface of an external tool or reference requires them (Sec. 8.3, linear-Gaussian check).

8.2 Main comparison: quality, model size, data, and baselines

One comparison, run identically on every task, opens the results. Table 3 trains the same architecture at three sizes and three simulation budgets for each reference task, evaluates every cell by median FID against that task's reference, and places the external tools in the same table at the shared budget. Figure 2 shows the corresponding posteriors themselves; Fig. 3 plots the grid.

Four questions, four answers.

1. *Is the quality acceptable?* At its best budget every task in the grid reads FID 0.01–0.05 — for scale, a 0.1 sd bias in each of d dimensions gives FID $0.01\,d$, so this is sub-tenth-of-a-sd accuracy — including 0.035 on the five-parameter Merton and 0.044 on pharmacokinetics. The posteriors of Fig. 2 carry the same message visually: location, scale, correlation, and ridge curvature are reproduced.

2. *Does model size buy accuracy?* It depends on the parameter dimension, and the grid separates the cases. At $d = 2$ the three columns differ by less than a factor of two, non-monotonically, and the smallest model is the best column at the largest budget — the exception is Fisher–KPP at the shortest budget (0.277 → 0.084 across the row), where 40 spatio-temporal observations are the

		tiny (0.1M)	small (0.5M, default)	big (2.3M)
GBM	20,000 sims	0.039	0.039	0.037
(exact conjugate;	120,000	0.016	0.021	0.024
200 sets)	480,000	0.010	0.012	0.018
	external tools	raw data 2.3–8.5; hand-derived inputs 0.34–1.5		
Ornstein–Uhlenbeck	20,000	0.091	0.070	0.079
(exact-lik. MCMC;	120,000	0.042	0.055	0.054
100 sets)	480,000	0.028	0.029	0.039
	external tools	1.2 (FMPE) – 2.7 (NPE)		
Fisher–KPP, $K = 40$	20,000	0.277	0.118	0.084
(PDE-lik. MCMC;	120,000	0.053	0.036	0.045
32 sets)	external tools	0.5 (NPE); 76 (FMPE)		
Merton, $d = 5$	120,000	0.074	0.052	0.051
(mixture-lik. MCMC;	480,000	MT480T	0.035	MT480B
32 sets)	external tools	— (random designs; fixed-length interfaces do not apply)		
pharmacokinetics, $d = 3$	120,000	0.063	0.044	0.038
(log-normal MCMC;	480,000	PT480T	0.070–0.147	PT480B
32 sets)	external tools	—		

Table 3: The main grid: median FID (over the stated test observation sets) of the same architecture at three sizes (columns) and increasing simulation budgets (rows), for every task with a validated reference; cells average over training seeds where several were run, and the big column uses the width-scaled learning rate. GBM, OU, and Fisher–KPP use the fixed-design protocol shared with the external tools, which enter the same table at the 20,000-simulation budget with their package defaults (ranges over `sbi` NPE/FMPE, BayesFlow 2, and the Simformer port; Sec. 8.3 attributes the failures). Merton and pharmacokinetics are trained and evaluated in the random-design mode on their fixed evaluation batteries, scored against the converged references of Sec. 6.2 (two-chain floors 0.0065 and 0.0023); their 480,000 pharmacokinetic entry is a range over two independently trained models — the budget reversal of Sec. 8.4. External tools consume fixed-length vectors and have no entry on the two random-design systems.

richest conditioning in the grid. At $d = 3$ and $d = 5$ width does buy accuracy, monotonically and by 1.4–1.7× over the same 24× parameter range, but it saturates: on Merton the two largest models differ by less than the reference's own floor. And where width does help, data helps more: on Merton the 4× data step at fixed width (0.052 → 0.035) beats the 24× parameter step at fixed data (0.074 → 0.051). The five-size ladder of Table 10 places the saturation point: below 10^4 parameters at $d = 2$, and between 10^5 and 5×10^5 at $d = 3$ and $d = 5$, where the last doubling of width buys 1.16× (pharmacokinetics) and nothing measurable (Merton).

3. *Does data buy accuracy?* Every column of every task improves monotonically down the rows, with no saturation anywhere in the measured range (Fig. 3), and the 4× data step moves every task more than the 24× parameter step does — with one measured exception, pharmacokinetics, where the largest budget is worse and the cause is localized to a single parameter (Sec. 8.4). Data, not capacity, is the axis worth paying for — and Sec. 8.4 shows the budget must be spent on *fresh* simulations, not on passes over stored ones.

4. *Are the hyperparameters tuned?* One coupling matters: the learning rate must shrink with width, or instability reads as a capacity effect (GBM big model: median FID 0.068 at the default 3×10^{-4}, 0.024 at 10^{-4}; the grid reports the stable setting). Everything else is the package default on every task; the solver control of Sec. 8.4 and the seed ranges of Fig. 3 bound what tuning could add.

At the shared 20,000-simulation budget, every column of the grid beats every external tool on every task it enters — by one to two orders of magnitude on GBM and OU on raw data, and by 2–6× on Fisher–KPP — and the gap survives hand-engineering the external tools' inputs; Sec. 8.3 dissects where it comes from.

Figure 2 shows the object the package produces — the joint posterior itself. For every system with an exact or validated reference, the posterior of the trained model is overlaid on the reference for one representative test observation set — three systems with simple posteriors that serve as checks (linear-Gaussian, GBM, OU), and three whose posterior geometry is nontrivial: the Merton jump-diffusion, where volatility and jump rate compete to explain the same variance; pharmacokinetics at six irregular

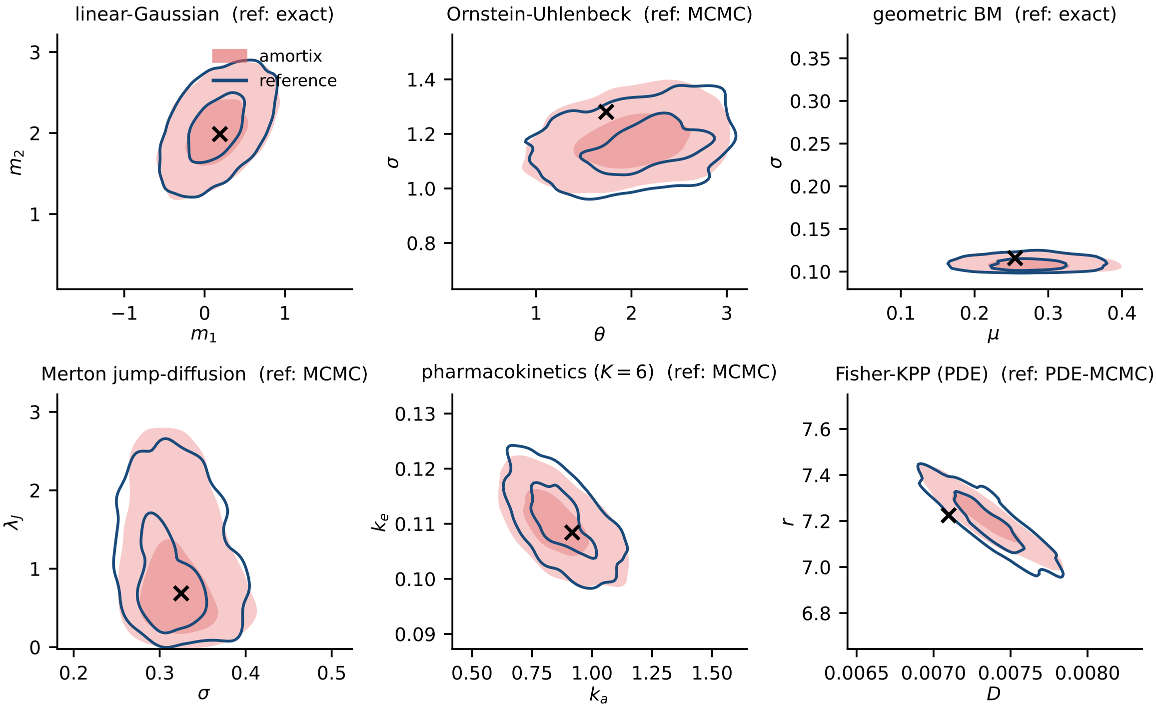

Figure 2: Joint posteriors of `amortix` (filled contours: 39% and 86% highest-density regions) against exact or validated-MCMC reference posteriors (open contours, same levels) on one test observation set per system; crosses mark the true parameters. Top row: systems with closed-form or cheap exact references. Bottom row: systems where the reference itself required exact-likelihood MCMC with two-chain validation — including the ridge-shaped Fisher–KPP posterior, where the model reproduces the curvature, not just the moments.

blood draws, the data-poor clinical regime; and Fisher–KPP, whose posterior concentrates on the curved ridge $Dr \approx$ const fixed by the observable front speed. The learned posteriors reproduce location, scale, orientation, and — where present — the curvature of the ridge.

Table 4 reports the corresponding global metric on the full test batteries, and Table 5 the calibration screen for the nine systems with no validated reference. Every reference row is evaluated in the package's default mode — one network per system, a fresh random design per test observation set — and reads median FID 0.01–0.07 on five of the six systems, the five-parameter Merton jump-diffusion included once its reference is computed with converged chains. The single elevated entry is pharmacokinetics (0.13), whose posterior at dense designs contracts to within a few multiples of the network's absolute error floor (Sec. 8.5). On the screen systems, 34 of 36 parameters pass, with the marginal cases moving between retrainings as expected at the screen's resolution limit. Per-parameter p-values, rank distributions, and coverage curves are in Appendix A.

Pharmacokinetics is the exception, and its mechanism is dissected in Sec. 8.5: its posterior is the narrowest in the suite, so the network's absolute error — comparable to every other system's — reads as a large FID once divided by that width, and the same narrowness makes it the one system where more training simulations do not help (Sec. 8.4). Together with Table 4, this figure is the central evidence that one small default model recovers full posteriors across the suite.

8.3 Comparison with existing tools

The headline numbers of this comparison already appear inside Table 3; this subsection specifies the protocol and attributes the failures to their mechanisms.

Protocol. We compare against the directly comparable amortized estimators available in maintained packages: NPE with its default masked autoregressive flow (Papamakarios et al., 2017) and FMPE (Wildberger et al., 2023) from `sbi` 0.27 (Boelts et al., 2024); the coupling-flow and flow-matching networks of BayesFlow 2.0 (Radev et al., 2020); and, because it is the closest architecture to ours in the literature, a port of the Simformer (Gloeckler et al., 2024) built from its authors' minimal example (transformer–

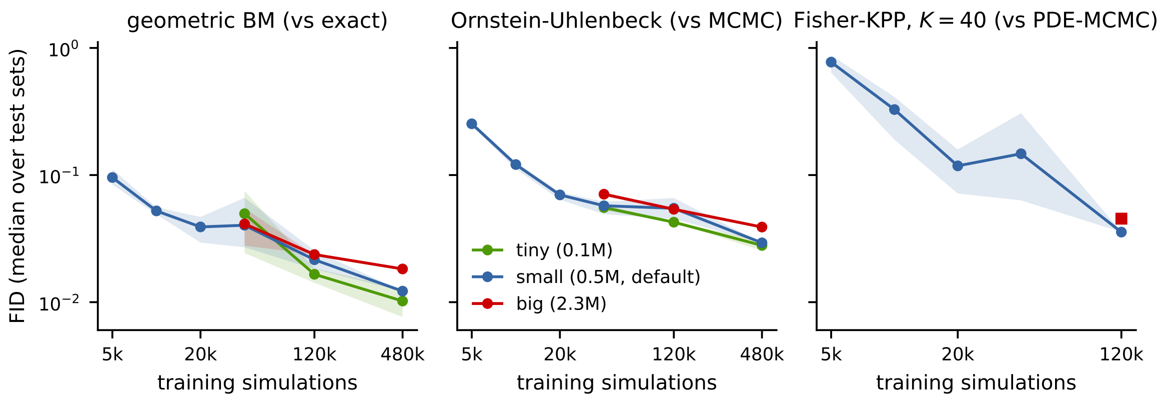

Figure 3: Median FID versus simulation budget at three model sizes (both axes logarithmic; optimizer steps scale with the budget at constant epochs; the big model uses the width-scaled learning rate). Lines: seed means; bands: seed ranges where multiple seeds were run. The widening Fisher–KPP band at intermediate budgets is seed-level freedom along the weakly identified ridge $\ln(r/D)$, which contributes little to the loss; refining the sampling ODE solver from 20 midpoint steps to 60 RK4 steps changes nothing measurable. On the two-parameter systems (left, middle) the three sizes coincide within seed spread — the budget axis, not the parameter axis, buys accuracy. Fisher–KPP (right): default size, with the big model at 120,000 marked by the square. References: exact conjugate posterior (GBM), exact-likelihood MCMC (OU), PDE-likelihood MCMC (Fisher–KPP, fixed $K = 40$ design).

system	d	reference	random designs, fresh per set	FID
geometric Brownian motion	2	conjugate, per design	$K = 2$–100 (600)	0.01
linear-Gaussian	4	exact subset posterior	$K = 2$–6 (300)	0.03
Ornstein–Uhlenbeck	2	exact-likelihood MCMC	$K = 2$–100 (600)	0.05
Fisher–KPP (PDE)	2	PDE-likelihood MCMC	$K = 10$–90 (96)	0.07
Merton jump-diffusion	5	Poisson-mixture MCMC	$K = 5$–100 (96)	0.05
pharmacokinetics	3	log-normal MCMC	$K = 5$–50 (96)	0.13

Table 4: Systems with reference posteriors: median FID (10) over the design-size buckets of Table 11, for the best trained model of the default architecture on each system. Every row is evaluated in the package's default mode: each test observation set (count in parentheses) draws a fresh random design of the stated size range, and the reference posterior is recomputed on exactly those points. Per-K breakdown: Table 11. Every reference here passes the two-chain check of Sec. 6.2 at 2×10^5 draws; against the 2,000-draw chains used in an earlier version of this table the Merton and pharmacokinetics rows read 0.19 and 0.18, at or below the sampler's own floor. Pharmacokinetics is elevated because its posterior at dense designs contracts to within a few multiples of the network's absolute error floor (Sec. 8.5).

diffusion over the joint, random condition masks, their architecture and SDE settings, full 75,000-step training). All arms receive the same simulation budget (20,000 draws from the same prior and simulator), run with their package defaults and their own stopping rules, and are compared on exactly three axes: training cost (wall-clock time to fit at this budget), accuracy (posterior-mean bias and width ratio per parameter against the reference posterior, as in Sec. 6.2), and inference cost (wall-clock time per observation set for 2,000 posterior samples). Non-amortized inference sets the scale on both ends: on GBM, whose likelihood is a three-line formula, adaptive Metropolis costs 42 ms per observation set and nothing amortized is worth its training bill for a single instance; on Fisher–KPP, one likelihood evaluation is one PDE solve, so the same reference sampler spends ≈12,000 solves — minutes of computation — on *each* observation set; and for Heston or Hodgkin–Huxley no likelihood formula exists at any price. Because these estimators consume fixed-length vectors, the comparison uses fixed observation designs — the only experiments in this report, apart from the linear-Gaussian check, run in that mode: a fixed design is the interface the external tools require, not the mode this package is built for (Sec. 8.5). Three systems are used, chosen to separate the possible explanations of any performance gap: a multiplicative process where input conditioning is hostile (GBM, exact conjugate reference), an additive process where it is comparatively benign (OU, exact-likelihood MCMC reference, two validated chains per observation set), and a PDE where every likelihood evaluation requires a forward solve (Fisher–KPP), so that the cost relationship between reference and amortized inference is inverted. Where informative, external arms are

system	d	SBC pass
Cox–Ingersoll–Ross	3	3/3
double well	3	3/3
stochastic Lotka–Volterra	4	3/4
FitzHugh–Nagumo	4	4/4
SEIR	5	5/5
polynomial-drift SDE	5	5/5
Heston	5	5/5
Hénon–Heiles	3	3/3
Hodgkin–Huxley	4	3/4

Table 5: Systems without a validated reference (no tractable likelihood, or an MCMC reference that fails the two-chain check of Sec. 6.2): parameters of the default model passing the SBC uniformity screen ($p > 0.05$, 500 observation sets $\times$ 200 draws; random-design systems over mixed designs). Individual marginal cases move between retrainings, as expected at the screen's resolution limit (Sec. 6).

method	input	train	inference	FID
`amortix` CFM	raw prices	236 s	70 ms	0.04
`sbi` NPE	raw prices	170 s	36 ms	4.8
`sbi` FMPE	raw prices	215 s	359 ms	3.8
BayesFlow 2 coupling	raw prices	1252 s	0.3 ms	2.6
BayesFlow 2 flow matching	raw prices	790 s	95 ms	2.3
Simformer port	raw prices	703 s	1049 ms	8.5
`sbi` NPE	log-prices	157 s	25 ms	5.8
`sbi` NPE	log-returns	52 s	26 ms	1.5
`sbi` FMPE	log-returns	99 s	378 ms	0.8
BayesFlow 2 coupling	log-returns	1254 s	0.3 ms	1.2
BayesFlow 2 flow matching	log-returns	771 s	97 ms	1.4
Simformer port	log-returns	586 s	1053 ms	0.34
exact-likelihood MCMC	log-returns (analytic)	—	42 ms (CPU)	reference

Table 6: GBM: all methods at package defaults, one H200 GPU, equal 20,000-simulation budget, 200 test observation sets against the exact conjugate posterior. FID is the Fréchet distance (10) to the reference, median over test observation sets (the distribution across sets is right-skewed; means are 1.5–3$\times$ larger, per-arm values in `results/`). Middle block: raw data, the amortized default regime. Bottom block: the same external tools given hand-derived input coordinates. Inference time is per observation set for 2,000 samples.

additionally given hand-derived inputs — log-prices, or per-gap log-returns, the exact coordinates of the GBM likelihood factorization — to separate the effect of the input representation from the generative engine.

Multiplicative process (GBM). Table 6 compares all methods on one device and one metric. At package defaults on raw price data, every external configuration misses the posterior by one to two orders of magnitude more than `amortix`: FID 2.3–4.8 against 0.04. The dominant mechanism is the input representation, not the generative engine (Sec. 9); `sbi` itself emits a warning about precision loss during z-scoring on this data. Hand-derived coordinates (bottom block of the table) help but do not converge to a common answer: with identical log-return inputs the external tools land at median FID 0.34–1.5 — the best of them still an order of magnitude above `amortix` operating on raw data — and they fail in different directions: NPE remains over-dispersed (σ width 1.44), both BayesFlow arms become overconfident (widths 0.42–0.85, the silently dangerous direction), and the Simformer port, best by median, has the heaviest tail across observation sets (mean FID 1.9, five times its median). Log-prices alone do not help NPE at all (5.8 vs 4.8 raw). At fixed input coordinates the flow-matching engine beats the autoregressive flow (0.8 vs 1.5 on log-returns), consistent with the engine choice of Sec. 3. Every defect here is invisible without an exact reference and lies within or near the blind region of SBC at practical sizes (Sec. 6); the failure directions replicate across devices and seeds (per-parameter decompositions for an independent laptop-CPU run of every arm are in `results/`).

method (raw values)	train	inference	FID
`amortix` CFM	226 s	77 ms	0.07
`sbi` NPE	134 s	19 ms	2.7
`sbi` FMPE	114 s	279 ms	1.2
exact-likelihood MCMC	—	124 ms (CPU)	reference

Table 7: Ornstein–Uhlenbeck (additive control), fixed 74-point design, 20,000-simulation budget, 100 test observation sets, one H200 GPU. FID: median Fréchet distance to the exact-likelihood MCMC reference (two validated chains per observation set).

method (40 observed values)	train	inference	FID
`amortix` CFM	591 s	69 ms	0.07
`sbi` NPE	235 s	19 ms	0.5
`sbi` FMPE	163 s	703 ms	76
PDE-likelihood MCMC	—	271 s (CPU)	reference

Table 8: Fisher–KPP, fixed $K = 40$ spatio-temporal design, 20,000-simulation budget, 32 test observation sets, one H200 GPU. FID: median Fréchet distance to the PDE-likelihood MCMC reference ($\approx$ 12,000 PDE solves per observation set; the amortized model's entire training equals the reference cost of fewer than three observation sets). The FMPE value is its diffusivity failure, discussed in the text; the defect is invisible to calibration screens at practical sizes.

Additive control (Ornstein–Uhlenbeck). OU is an additive process, so the floating-scale problem of raw prices is absent by construction, and running the identical protocol on it shows how much of the GBM failure that problem accounts for (Table 7). The external tools improve substantially: their volatility bias becomes consistent with zero ($+0.19 \pm 0.19$ sd for NPE, $+0.08 \pm 0.12$ for FMPE), and their width inflation drops from 2.5–2.6 to 1.9 (NPE) and 1.6 (FMPE). It does not disappear: extracting quadratic-variation information from 74 raw values remains a representation task that a z-scored vector input leaves entirely to the network. `amortix` reads 1.03 and 1.05 on the same observation sets.

Simulator-limited regime (Fisher–KPP). The third system inverts the cost structure of the first two: every likelihood evaluation requires solving the forward model. A one-dimensional reaction–diffusion equation is inexpensive in absolute terms (16 ms per solve here); what makes the case informative is the *ratio* between reference and amortized inference, because both sides scale linearly with the solve cost — a chain of n likelihood evaluations costs n solves per observation set whatever the forward model, while the amortized cost per observation set does not involve the simulator at all. The measured ratio below transfers to genuinely expensive simulators, where the same reference sampler would take days per observation set. A single fixed design of $K = 40$ random (time, sensor) points is shared by all observation sets and methods; the reference is the PDE-likelihood sampler of Sec. 6.2 with two validated chains per observation set (271 s per observation set for the pair; two-chain validation over 32 observation sets: worst mean discrepancy 0.22 sd, mean 0.07 sd). Table 8 gives the results. `amortix` matches the reference at FID 0.07, answering in 69 ms per observation set, and its entire training cost (591 s) equals the reference cost of fewer than three observation sets — amortization pays for itself almost immediately in any repeated-inference setting. NPE shows moderate error (FID 0.5, from widths of 1.26–1.28); FMPE at defaults fails outright on the diffusivity ($+7.3 \pm 2.7$ sd bias at $12.5\times$ width, FID 76) while remaining reasonable on the reaction rate — default configurations behave discontinuously across parameters, detectable here only because the expensive reference was computed.

Scope of the comparison. Every external arm ran at its package defaults under our simulation budget; specialists could improve each with problem-specific embedding networks or summary statistics. The comparison is designed to measure exactly that dependence: what accuracy is delivered when the input representation is not hand-engineered per problem. The Simformer arm in particular is a port of a minimal example rather than a tuned configuration, and its numbers should be read as representative of that recipe, not of the method's ceiling. Conversely, the variable-design capability of Sec. 8.5 is not exercised here at all, since the external estimators consume fixed-length inputs by default.

		FID at epochs			
system	trajectories	10	35	120	120 + subsampling
GBM	10,000	0.105	0.059	0.965	0.114
	30,000	0.036	0.047	0.659	—
	120,000	0.044	0.026	0.131	0.277
OU	10,000	0.206	0.151	1.892	0.253
	30,000	0.090	0.095	1.648	—
	120,000	0.038	0.066	0.648	0.143

Table 9: Median FID when the same fixed-design training is given more simulated trajectories (rows) or more passes over them (columns), at the default model size; cells on an anti-diagonal cost equal compute. The last column repeats the 120-epoch run with a fresh random 75% token subset at every optimizer step.

8.4 Data reuse and the capacity ladder

The grid of Table 3 scales optimizer steps with the simulation count at constant epochs. Two further grids separate what a budget buys when that convention is broken: passes over stored simulations versus fresh ones, and parameters versus data at five model sizes.

Fresh simulations beat repeated passes. Table 9 varies the number of simulated trajectories and the number of epochs independently, for the fixed-design GBM and OU models at the default size. Two readings. Along constant compute — 10,000 trajectories $\times$ 120 epochs versus $120{,}000 \times 10$, equal optimizer steps — fresh data wins by an order of magnitude or more in FID. And passes are not merely diminishing but destructive: at 120 epochs every configuration is worse than at 35, including the largest training set (OU: $0.066 \to 0.648$). The mechanism is memorization of the tokens: virtual times and reference noise are redrawn at every step (Sec. 5), so what repeats across passes is only the fixed observation vectors. Consistent with this, redrawing a random 75% token subset at every optimizer step (`token_dropout`) largely repairs the 120-epoch column — OU $1.89 \to 0.25$ at 10,000 trajectories, $0.65 \to 0.14$ at 120,000 — while slightly hurting the one cell that had barely overfitted (GBM at 120,000: $0.13 \to 0.28$): training on subsets while evaluating on the full design is its own train/test mismatch, so subsampling is a repair, not a free improvement. The design-amortized procedure performs the same augmentation natively and without the mismatch — every optimizer step sees a fresh design drawn from the law used at evaluation — which is why the amortized network of Table 11 is at least as accurate as its fixed-design counterpart at the same budget.

The floor is statistical, not an optimizer artifact. An error that moves with neither data nor capacity invites the hypothesis that it is the optimizer: a first-order method with a decaying step leaves the iterate jittering in a noise ball, and second-order information should shrink it. The conditional flow-matching objective is a nonlinear least-squares problem, so its generalized Gauss–Newton matrix is exactly $J^\top J$ for the residual $v_\theta - (m_1 - m_0)$, and that matrix can be applied without ever forming it. We ran inexact stochastic Gauss–Newton — truncated conjugate gradients on $(J^\top J/N + \lambda I)\delta = -J^\top r/N$ with a warm start from the previous direction, a backtracking step, and Levenberg damping adapted from the accepted step — continuing from the trained model, against an Adam control given the same wall-clock from the same starting point. It loses by an order of magnitude: from a starting FID of 0.010, Gauss–Newton reaches 0.066 after 1,500 steps while the Adam control reaches 0.007; solving the local system more accurately (20 CG iterations instead of 5) makes it worse still (0.23), and from a random initialization it reaches 0.43 where Adam reaches 0.033 in the same time.

The failure is diagnostic rather than incidental. The damping collapses to its floor, full steps are accepted, and the minibatch loss does decrease — while held-out accuracy degrades, the signature of fitting the batch. The arithmetic explains it: a batch of 4,096 observation sets on a two-parameter problem contributes 8,192 residual equations against 4.8×10^5 parameters, so the local system is underdetermined by two orders of magnitude and its minimum-norm solution interpolates that batch's noise — which is why solving it more accurately hurts. Raising the effective batch to 65,536 by accumulating the products over chunks confirms the mechanism (from a random start, FID 1.8 after 150 steps against $\approx$6 for the unaccumulated version at the same step count) without changing the conclusion. Underneath sits the reason no batch size we can afford would rescue it: the regression target of (3) has an irreducible conditional variance — the residual sits at 0.77 in normalized units while a step buys 0.005 — so curvature

model (width × blocks)	params	GBM $d = 2$	OU $d = 2$	pharmacokin. $d = 3$	Merton $d = 5$
pico (8 × 2)	7,114	0.028–0.029	0.052–0.095	0.675	0.243
nano (16 × 2)	25,730	0.021–0.027	0.042–0.054	0.167	0.091
tiny (32 × 2)	97,522	0.014–0.018	0.042–0.043	0.063	0.074
small (64 × 3, default)	483,346	0.018–0.026	0.045–0.066	0.044	0.052
big (128 × 4)	2,321,554	0.024	0.054	0.038	0.051

Table 10: Capacity ladder: median FID at a fixed budget of 120,000 simulations; sizes are given as width × transformer blocks, with parameter counts for the GBM instance (counts vary by a few percent across systems with the input dimension). Ranges span independent training seeds. GBM and OU use the fixed-design protocol of Table 9; pharmacokinetics and Merton use the design-amortized protocol, evaluated against the converged references of Sec. 6.2 whose own two-chain floors are 0.0023 and 0.0065 — every entry in those two columns is above its floor by a factor of 8 or more. The big row uses the width-rescaled learning rate (see text). The ladder is flat at $d = 2$, develops a floor near 10^5 parameters at $d = 3$, and is monotone through the full range at $d = 5$.

is estimated from data in which the signal is a fraction of a percent of the target noise. The small steps and weight averaging of the first-order recipe are not a limitation to be overcome here; they are what converts that noise into an estimate of the conditional mean.

The capacity floor moves with the parameter dimension. Table 10 trains the same architecture at five sizes spanning 7 thousand to 2.3 million parameters, at a fixed budget of 120,000 simulations, on four systems of increasing parameter dimension. At $d = 2$ the ladder is flat: a 7,000-parameter model, 70× smaller than the default, matches the default within seed noise on GBM. At $d = 3$ the floor is visible: below $\sim 10^5$ parameters accuracy degrades several-fold. At $d = 3$ the floor sits between the two smallest sizes: accuracy improves 2.7× from nano to tiny (0.167 → 0.063) and only 1.6× over the twenty-four-fold step from tiny to big (0.063 → 0.038). At $d = 5$ the ladder is monotone until it saturates at the default size: 0.243, 0.091, 0.074, 0.052, 0.051 from pico to big, so every halving of width below the default costs accuracy while doubling it beyond buys nothing measurable — the two largest entries differ by less than the reference's own floor of 0.0065. The capacity floor therefore rises with the parameter dimension, and above it the simulation axis takes over again: at 480,000 simulations the default Merton model reaches 0.035. Two couplings matter when reading such ladders. The learning rate must shrink with width: at width 128 the default 3×10^{-4} is unstable and masquerades as "bigger is worse" (GBM median FID 0.068), while halving it restores parity (0.024); the big row of the table reports the stable setting. And adding parameters without adding data buys nothing here, whereas adding data does: at 480,000 simulations the default model reaches FID 0.012 (GBM) and 0.029 (OU), while the big model at the same enlarged budget and rescaled learning rate reads 0.018 and 0.039. The same lever closes the capacity gap at $d = 5$: quadrupling the simulations improves the default Merton model from 0.213 to 0.186 and the big one from 0.206 to 0.200, so once data is scaled the default *overtakes* the big model — capacity binds at a fixed simulation budget, but the data axis ultimately outbids the parameter axis everywhere we measured. The one exception in the suite is pharmacokinetics, where the budget axis does not merely saturate but reverses: on its converged $K = 6$ battery the default model reads 0.044 at 120,000 simulations and 0.070 and 0.147 for two independently trained models at 480,000. The degradation is localized. Posterior widths stay within 5% of the reference for every model and both other parameters are unaffected — the entire effect is the elimination rate k_e, whose posterior-mean error grows from 0.08 to 0.26 and 0.37 reference standard deviations, and whose posterior is the narrowest in the suite. It is reproducible across the two long runs, so it is not seed noise, and we report it as unexplained: it is the one place in this report where more simulations make an entry worse, and it happens to the parameter that the precision floor of Sec. 8.5 squeezes hardest. On low-dimensional systems the parameter axis saturates by a few thousand weights; the simulation axis did not saturate on any other system.

Costs, for orientation. The saturated models of Tables 4 and 5 train in 19–63 minutes per system on one H200 GPU (120,000 simulations; 66–72k optimizer steps); at the 20,000-simulation budget of the comparisons above, training takes 6–12 minutes. On a laptop CPU, training takes roughly four times longer and produces bit-identical posteriors, because all stochastic generation is kept on a CPU random number generator by design. Inference draws 2,000 posterior samples in 426 ms per observation set on the laptop CPU and 72 ms batched (135 ms single) on the GPU. The small external models of Sec. 8.3 train and sample faster at their defaults; the tiny-model regime is kernel-launch-bound on GPUs (`sbi`'s default flow trains 6× slower there than on the laptop CPU).

system	FID at design size K (median)			specialist
	sparse	medium	dense	at its own K
geometric Brownian motion	0.006 (5)	0.008 (20)	0.018 (100)	0.012 (95)
Ornstein–Uhlenbeck	0.072 (5)	0.045 (20)	0.023 (100)	0.029 (74)
linear-Gaussian	0.012 (2)	0.035 (4)	0.046 (6)	0.039 (6)
Fisher–KPP	0.073 (10)	0.050 (40)	0.126 (90)	0.036 (40)
Merton jump-diffusion	0.027 (5)	0.046 (20)	0.057 (100)	—
pharmacokinetics	0.069 (5)	0.125 (20)	0.339 (50)	—

Table 11: One design-amortized network per system, evaluated at design sizes never fixed in advance (the bucket's K in parentheses): each test observation set draws a fresh random design and its reference posterior is recomputed on exactly those points. Last column: a separate network of the same architecture and simulation budget trained on a single fixed design, which exists only at its own K. GBM and OU are evaluated on the denser K-grid of Fig. 4. The Merton and pharmacokinetics rows use the converged references of Sec. 6.2; with the 2,000-draw chains of an earlier version they read 0.17–0.21 and 0.12–0.42, which the sampler's own floor (0.24–0.30 and 0.06–0.08) makes uninformative for Merton and distorts for pharmacokinetics. Pharmacokinetics rises steeply with K — the precision floor, audited in the text.

8.5 Amortization over observation designs

The package's central claim is that one trained network answers $p(m \mid S)$ for *every* observation design S — any number of points, any placement. Table 11 and Fig. 4 test the claim on every system of the suite where a per-design reference can be computed and validated — six of the fifteen: each test observation set draws a fresh random design, the reference posterior is recomputed on exactly those points, and FID is evaluated per design-size bucket. Two readings. First, wherever a fixed-design specialist of the same architecture and simulation budget exists, the single amortized network is comparable to it at the specialist's own design size (GBM 0.018 vs 0.012; OU 0.023 vs 0.029; linear-Gaussian 0.046 vs 0.039; Fisher–KPP 0.050 vs 0.036) and is the only one of the two that answers anywhere else on the K axis. Amortization over designs is not a trade-off against accuracy at any point we measured; it also removes a failure mode, since resampling the observation subset at every optimizer step doubles as data augmentation (Sec. 8.4 measures what the fixed-design procedure pays for reusing identical tokens across passes). Second, the two systems with elevated FID are not noise, and the two paragraphs after the table take them apart.

The amortized precision floor. Pharmacokinetics is the one system whose FID *rises* with the design size ($0.069 \to 0.339$ from $K = 5$ to 50 against converged references), and we audited the number before interpreting it: an independent exact posterior (Laplace mode plus importance sampling with 2×10^5 exact likelihood evaluations per observation set) agrees with the MCMC reference to 0.0–0.23 sd, and the model's FID against either reference coincides — the value is real, not a reference artifact. It is also not a budget artifact: a model trained on 4× the simulations and 3× the optimizer steps is not better on the same test sets — in fact worse, the budget reversal of Sec. 8.4. The mechanism is a race between two scales, both measured directly rather than inferred from FID. Table 12 reports, per parameter, the absolute error of the model's posterior mean and the width of the reference posterior, both in units of the prior range. The absolute error stays at the few-$\times 10^{-3}$ level on both systems — 3–6×10^{-3} on GBM, 1–8×10^{-3} on pharmacokinetics — although their posterior widths differ by a factor of thirty: it is a property of the trained network, not of the task. What differs is the denominator FID divides by. On the pharmacokinetic model every parameter's error is 11–15% of its posterior standard deviation, against 2–5% on GBM, and the resulting FID (0.35 at $K = 50$) splits roughly evenly between the posterior-mean shift and a covariance-shape term — it is not a single bad parameter but the whole posterior being resolved to a scale finer than the network's floor. The stochastic systems never enter this regime, because diffusion noise keeps their posteriors an order of magnitude wider than the error floor even at $K = 100$; Fisher–KPP approaches it only at its densest bucket (0.126 at $K = 90$). This is also a caveat on FID itself: as a scale-relative metric it reports a model that is uniformly accurate in absolute terms as increasingly poor wherever the posterior it is compared against is narrow.

Where references end. Two further systems — Hénon–Heiles and Hodgkin–Huxley — are deterministic flows with Gaussian observation noise, so their likelihood is tractable in principle and a per-design reference looks within reach. It is not, and the reason is worth recording. Under deterministic chaos the posterior concentrates on needle-thin sets (Hénon–Heiles at $K = 20$: the ω_1 posterior has a standard

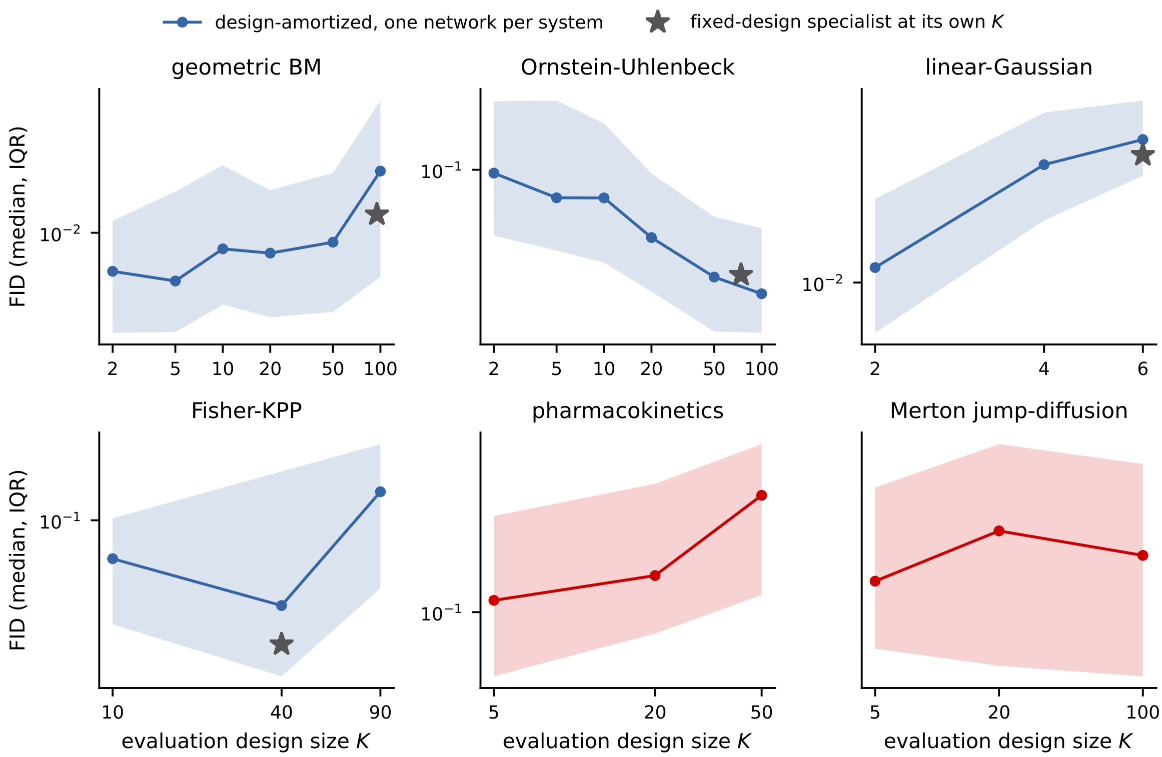

Figure 4: FID as a function of the evaluation design size, for one design-amortized network per system: median (line) and interquartile range (band) over the test observation sets of each bucket (100 per bucket for GBM, OU, and linear-Gaussian, 32 for the MCMC-referenced systems), each set with a fresh random design and a per-design reference. Stars: fixed-design networks of the same architecture and simulation budget, which exist only at their own K. Both axes logarithmic; note the different vertical scales.

deviation near 10^{-3} of the prior range while ω_2 stays prior-wide), and adaptive random-walk chains do not converge on such geometry: independent chains land 67–108 sd apart on half of the Hénon–Heiles test sets, and 1.8–18 sd apart on Hodgkin–Huxley, against the 0.22 sd worst case of every validated reference in this report. By the validation standard of Sec. 6.2 these references are rejected, and both systems remain under the calibration screen of Table 5. The precision-floor argument above predicts that a valid reference, if computed by a stronger sampler, would find elevated FID here as well.

Variable-design zoo. With the class rule and training procedure fixed, six further systems were trained without per-system adjustments. Heston (five parameters, latent volatility): all 25 SBC cases pass, and increment tokens give no advantage over point tokens, consistent with the class boundary (Sec. 4.3). Merton jump-diffusion: 25 of 25 cases after the heavy-tail treatment of Sec. 9, including dense-design cases that failed at $p < 10^{-3}$ before it; the residual width sits on the budget curve below. Hénon–Heiles (parameters of the potential of Lubich et al. 2015, observing one coordinate at random times): calibrated at every design size down to $K = 6$, where the posterior widens honestly rather than degrading. Hodgkin–Huxley (Hodgkin and Huxley, 1952): calibrated on mixed designs; one dense-design case shows the budget signature. Pharmacokinetics: at $K = 6$ irregular blood draws, bias -0.01 to -0.06 sd and widths 1.09–1.33 against exact-likelihood MCMC. Fisher–KPP: the posterior concentrates on the ridge $D \cdot r \approx$ const; the residual along-ridge bias decays with budget (Sec. 8.4).

Which encoder components make design amortization work is an architecture question, separate from the accuracy claims above; the controlled comparison of eight encoder variants that fixed the design of Sec. 4 is in Appendix B.

9 Failure analysis: conditioning of the encoder input

Every accuracy failure encountered during the development of this package — including, we argue, the external failures of Table 6 — reduces to a single cause appearing in three forms: the conditioning of

system	parameter	posterior sd	model error	error / sd
GBM, $K = 20$	drift μ	0.177	0.0034	0.02
	volatility σ	0.110	0.0058	0.05
pharmacokinetics, $K = 50$	absorption k_a	0.073	0.0083	0.11
	elimination k_e	0.0065	0.0010	0.15
	volume V	0.021	0.0031	0.15

Table 12: The precision floor, measured directly against the converged references of Sec. 6.2 for the best model of each system. Both quantitative columns are in units of the prior range: *posterior sd* is the width of the reference posterior, *model error* is $|\mathbb{E}_{\text{model}}[m] - \mathbb{E}_{\text{ref}}[m]|$, medians over 32 test observation sets with 4,000 samples per posterior. The model's absolute error stays at the few-$\times 10^{-3}$ level on both systems while the posterior widths differ by 30×, so the last column — the error a scale-relative metric sees — is 3–8× larger on the narrow pharmacokinetic posteriors than on the wide GBM ones.

the numbers presented to the set encoder. We document the three forms with their measurements and repairs; all three repairs are learnable modules rather than problem-specific transformations.

Form 1: floating multiplicative scale. On GBM, raw price tokens force the network to divide squared increments by squared levels internally; with global standardization this manifests as a volatility bias of +0.48 posterior sd (confirmed against the closed-form posterior) that no amount of training removes. The repair is the monotone mixture reparametrization of Sec. 4.3, which learns the log coordinate; the Yeo–Johnson ablation there shows why the parametrization, not the hypothesis class, is the operative choice. External estimators exhibit the same form on the same data (Table 6, raw-price rows).

Form 2: cardinality blindness. Normalized attention erases set size, which controls posterior width under design amortization; the probe measurements and the $\log K$/register-token repairs are given in Sec. 4.3, with the arm-by-arm consequences in Table 14.

Form 3: heavy tails. On the Merton jump-diffusion, squared increments have sample standard deviation 3.1×10^4 against a diffusion signal of order 10^{-2}–1; running feature normalization never converges, and the flow trains against a drifting representation, producing a +1.12 sd volatility bias. Encoder probing (Sec. 6.3) shows the information is extracted — the failing configuration encodes robust volatility statistics better than the healthy one — so the repair targets representation stability: logarithmic compression of increment features (near-identity on clean diffusions since $\log(1 + x) \approx x$ for small x) restores all 25 evaluation cases. A precision follow-up against the Poisson-mixture reference then isolated a smaller (+0.27 sd) residual whose mechanism is a jump-classification threshold that floats with the unknown volatility; a per-observation-set set-conditioning module (pooled statistics modulating feature channels, initialized to the identity) removes it as effectively as a hand-crafted per-observation-set rescaling ($+0.127 \pm 0.055$ vs $+0.094 \pm 0.054$ sd, statistically indistinguishable). This was one of three such paired comparisons in the project (log coordinate, set size, robust scaling); in all three the learnable module matched the hand-crafted repair once initialized at the working configuration. We summarize the pattern as a design principle: the hand-crafted fix identifies the mechanism and the initialization; the learnable module placed there is what ships.

Three implementation conditions proved necessary for the set-conditioning module and are recorded here because each was found through a failure: the modulation must be zero-initialized (training must start from the working baseline); it must act after feature normalization, not before (the reverse ordering re-creates the representation drift it is meant to cure); and pooled set statistics must exclude padding positions (a masking error here made padded and packed representations of the same observation set inequivalent, detected only because SBC and the exact-reference probe disagreed).

10 Discussion and limitations

The evidence assembled here supports a specific claim: for dynamical-systems inverse problems, the input representation of an amortized posterior estimator is not a preprocessing detail but the dominant accuracy factor, and it can be made part of the model rather than part of the problem specification. The comparison of Sec. 8.3 shows the cost of the alternative — fixed representations with global standardization — across

system	per-parameter SBC p-values
linear-Gaussian	m_1: 0.06 m_2: 0.60 m_3: 0.38 m_4: 0.56
Ornstein–Uhlenbeck	θ: 0.28 σ: 0.77
SEIR	β_1: 0.33 β_2: 0.22 α: 0.61 γ_r: 0.34 γ_d: 0.53
geometric Brownian motion	μ: 0.26 σ: 0.44
Cox–Ingersoll–Ross	a: 0.65 b: **0.001** σ: 0.73
double well	θ_1: 0.78 θ_2: 0.54 σ: 0.88
stochastic Lotka–Volterra	α: 0.07 β: 0.69 δ: 0.08 γ: 0.35
FitzHugh–Nagumo	a: 0.19 b: 0.15 ϵ: 0.90 I: 0.63
polynomial-drift SDE	c_0: 0.06 c_1: 0.96 c_2: 0.46 c_3: 0.80 σ: 0.50

Table 13: Per-parameter SBC uniformity p-values for the fixed-design gallery run of the standard-budget gallery run (500 observation sets × 200 draws; $p > 0.05$ passes). The single failure is marked in bold; see Sec. 8.2 for its behavior across retrainings.

four generative engines, on a problem where the correct answer is known in closed form. The OU control separates the mechanisms: on additive data the bias of the external estimators disappears and their width inflation drops by roughly a factor of two but does not vanish, so the conditioning pathology accounts for the dominant part of the failure and a smaller representation-learning gap accounts for the rest.

Limitations. (i) The remaining inaccuracies of the suite decay as measured power laws in training compute (Fig. 3); we have not searched for methods that improve the exponents. (ii) The power limits of SBC imply that small residuals can only be certified where an exact or validated reference exists; systems without tractable likelihoods inherit the weaker screen, and carrying reference-equipped systems alongside them in one suite is our current mitigation. (iii) The comparison of Sec. 8.3 uses package defaults; it measures the out-of-the-box regime, not the ceiling of any method. (iv) Bayesian experimental design — the intended consumer of design amortization — is not yet demonstrated; the training procedure of Sec. 5 was built so that the required object $p(m \mid S)$ is already coherent across subsets S. (v) No real-data study with posterior predictive checks is included yet. (vi) The polynomial-drift system uses a dense coefficient prior; the scientifically interesting variant is a *sparse* prior — train on coefficient vectors with random supports and test whether the posterior recovers the support — which requires extending the normalization (2) to priors with atoms at zero, a planned extension of the package. (vii) All priors are uniform on boxes; the probit normalization (2) extends to any factorized prior with tractable CDFs, but correlated priors would require a different normalization.

A Calibration detail for the fixed-grid systems

Per-parameter SBC p-values for the standard-budget gallery run (40,000 simulations, the run of Figs. 5 and 6) are listed in Table 13; Fig. 5 shows empirical-vs-nominal coverage of the central credible intervals, and Fig. 6 the underlying rank distributions.

B Encoder ablations for design amortization

The ablations of this appendix track the posterior width component rather than FID: the mechanism under study — set-size blindness of normalized attention — expresses itself specifically as width error, and FID would mix it with unrelated deviations. Table 14 and Fig. 7 compare encoder variants for design amortization on GBM, trained at $K \sim$ log-uniform$[2, 128]$ random-time designs and evaluated against the exact per-design posterior (96 observation sets per bucket). Three observations organize the table. First, index-based positions combined with in-place subsampling during training (leaving kept tokens at spread positions while inference packs contiguously) produce biases of +1.1 to +3.3 posterior sd with no individually incorrect component — the positional pathology of Sec. 4.2; repacking or replacing positions with time phases removes the bias entirely. Second, all point-token variants share inflated widths at dense designs, in proportion to how poorly their pooled representation encodes $\log K$ (rightmost column; the probe of Sec. 4.3) — including two attention mechanisms specifically constructed to exploit continuous time (a learned per-head time-kernel logit bias in the spirit of relative-position methods, and edge-message attention carrying gap-modulated increments on all pairs). Third, pair tokens with time-phase attention, plus the training procedure of Sec. 5, bring every bucket to within 7% of the exact width.

encoder variant	σ width ratio at K 5	9	20	100	$R^2(\log K)$
point tokens, index positions, spread subsampling	0.94*	1.13*	1.42*	2.08*	—
point tokens, index positions, contiguous	1.05	1.17	1.13	1.73	0.957
point tokens, time-kernel attention bias	1.03	1.14	1.31	2.51	0.712
point tokens, edge-message attention	1.03	1.15	1.29	2.45	0.498
point tokens, time phases	1.01	1.10	1.14	1.95	—
+ explicit $\log K$ feature	1.01	1.05	1.08	1.49	—
pair tokens, time phases	1.02	1.02	1.02	1.27	0.985
pair tokens + full training procedure	1.06	1.06	1.05	1.07	—

Table 14: GBM with variable designs: posterior width for the volatility σ against the exact per-design posterior (96 observation sets per design-size bucket) and, where measured, the probe R^2 for $\log K$ from the pooled encoder output. The table tracks σ because it is the parameter whose posterior width is governed by the observation density — the drift μ is weakly informed at every K, and all variants recover its posterior within a few percent, so it does not discriminate between architectures. *This row also carries biases of +1.09 to +3.30 posterior sd (the positional train/test mismatch of Sec. 4.2); all other rows have biases consistent with zero.

C Reference posteriors

GBM conjugate posterior. With observations at indices $i_0 = 0 < i_1 < \dots$ of the fine grid, log-returns $r_k = \log(S_{i_{k+1}}/S_{i_k})$ over gaps τ_k are independent $\mathcal{N}\big((\mu - \sigma^2/2)\tau_k, \sigma^2\tau_k\big)$. In the coordinates $(b, \sigma^2) = (\mu - \sigma^2/2, \sigma^2)$ the model is a Gaussian location–scale family with sufficient statistics $T = \sum_k \tau_k$, $R_1 = \sum_k r_k$, and $SS_w = \sum_k r_k^2/\tau_k - R_1^2/T$; draws from the standard conjugate posterior ($\sigma^2 = SS_w/\chi^2_{n-1}$, $b \mid \sigma^2 \sim \mathcal{N}(R_1/T, \sigma^2/T)$) are transformed to (μ, σ) and importance-resampled to the uniform box prior with the appropriate Jacobian weight. The sampler was validated against exact-likelihood MCMC on the same designs: over 24 comparisons, worst posterior-mean discrepancy 0.099 sd, width ratios 0.91–1.03.

Adaptive Metropolis. Random-walk Metropolis on the box, proposal $x' = x + s\,C\,\xi$, $\xi \sim \mathcal{N}(0, I)$. During burn-in (default $\max(500, n/2)$ iterations) the scalar s is adapted toward a 0.25 acceptance rate in blocks of 50 with a decaying Robbins–Monro gain, and once sufficient burn-in states are collected the shape C is replaced by the Cholesky factor of the empirical covariance rescaled by $2.38/\sqrt{d}$ (Haario et al., 2001); adaptation is frozen for the retained phase. Out-of-box proposals are rejected, which samples the box-truncated target exactly since the proposal is symmetric. Likelihoods: per-gap Euler–Maruyama Gaussian transitions for OU (matching the generative process, plus the stationary density of X_0); the exact GBM log-return density; the Poisson-mixture transition density for Merton (truncated at the numerically negligible jump count); log-normal residuals around the Bateman curve; a Gaussian observation model around the deterministic PDE solve for Fisher–KPP. Every chain used as a reference is paired with a second chain from a well-separated start; between-chain posterior-mean discrepancies are reported alongside the results they certify (Sec. 8.3).

D Correspondence with package options

in this report	in the package
point tokens	`embed="wpoint"`
pair tokens + set conditioning	`embed="wfilm"`
warped values/increments (fixed designs)	`embed="wbasis"` / `"wdiff"`
attention phases from physical time	`rope="time"`
fresh designs each optimizer step	`fit(retokenize=...)` via `make_retokenizer()`
variable-design problem wrapper	`amortix.designs.DesignProblem`
benchmark systems	`amortix.problems` (`DESIGN_ZOO`)
calibration screen / reference machinery	`sbc_design` / `amortix.mcmc` + likelihood factories
comparison scripts (Sec. 8.3)	`examples/baseline_{npe,ou,kpp,bayesflow,simformer}.py`

All defaults named in this report are selected automatically by `FlowPosterior(problem)`; the non-default options exist as reproducible controls for the comparisons above.

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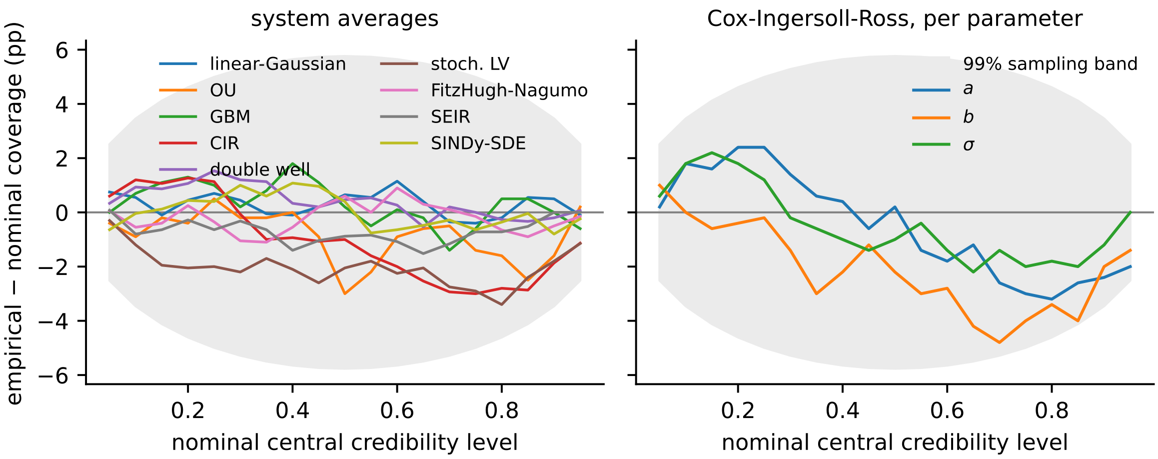

Figure 5: Empirical coverage of central credible intervals for the fixed-design gallery, shown as the deviation from the nominal level. For each system, 500 synthetic observation sets are simulated with known parameters and the posterior's central $q\%$ interval is checked against the truth; a calibrated posterior has zero deviation at every level, and the gray band is the 99% pointwise range induced by the finite number of test observation sets. Left: deviation averaged over each system's parameters. Right: individual parameters of Cox–Ingersoll–Ross, the system with the largest deviation in this run; the b curve reflects the ≈ 0.15 sd location shift discussed in the text. Rank-level detail for all 32 parameters is in Appendix A (Fig. 6).

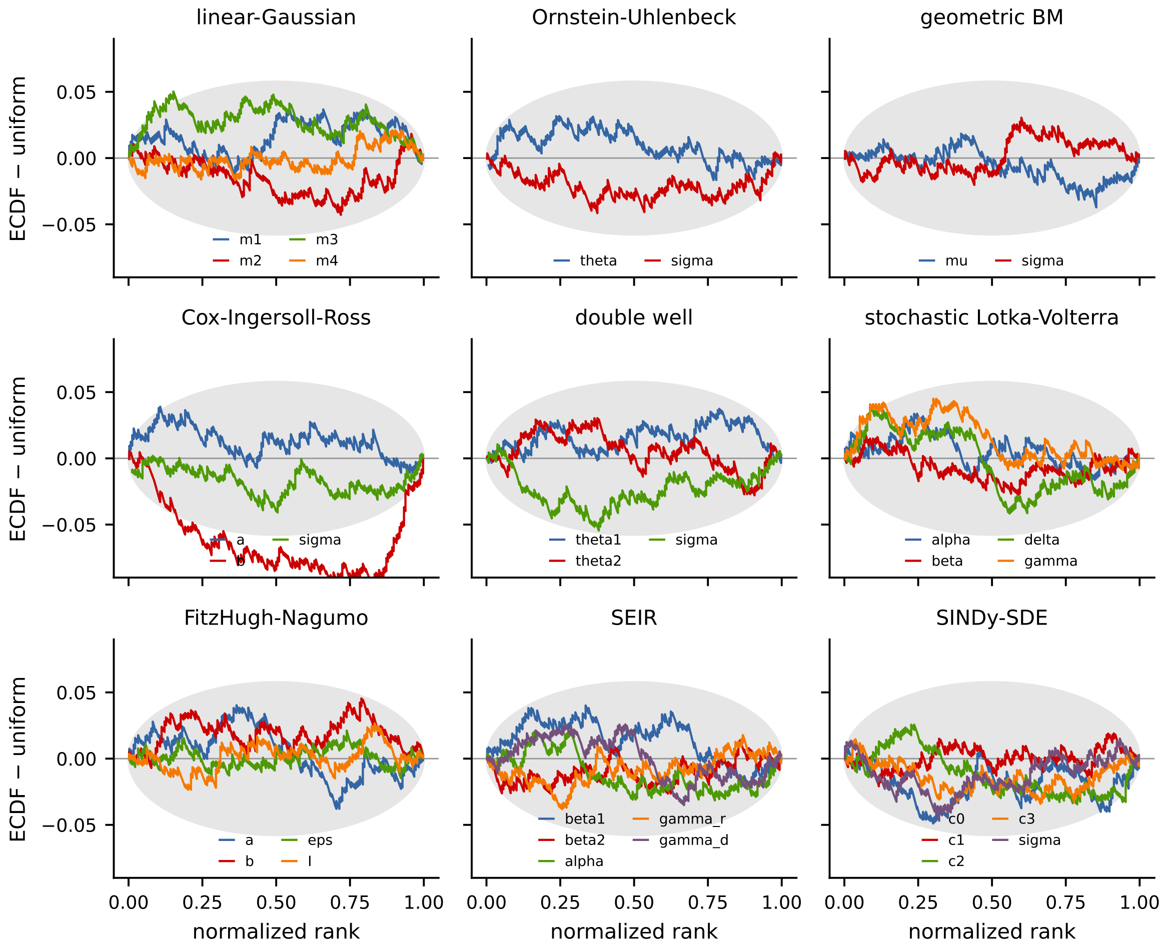

Figure 6: Calibration check for the fixed-design gallery. Procedure: for each system, 500 synthetic observation sets are simulated with known parameters; for each observation set the model produces 200 posterior draws, and the rank of the true parameter among the draws is recorded. If the posterior is calibrated, these ranks are uniform. Each curve shows, for one parameter, the deviation of the empirical CDF of normalized ranks from the uniform diagonal; the gray band is the 99% range attributable to sampling noise at 500 observation sets. A curve bulging below zero on the left indicates a posterior shifted low relative to the truth, and vice versa. All curves stay within the band except Cox–Ingersoll–Ross b, whose deviation corresponds to the ≈ 0.15 sd location shift discussed in the text.

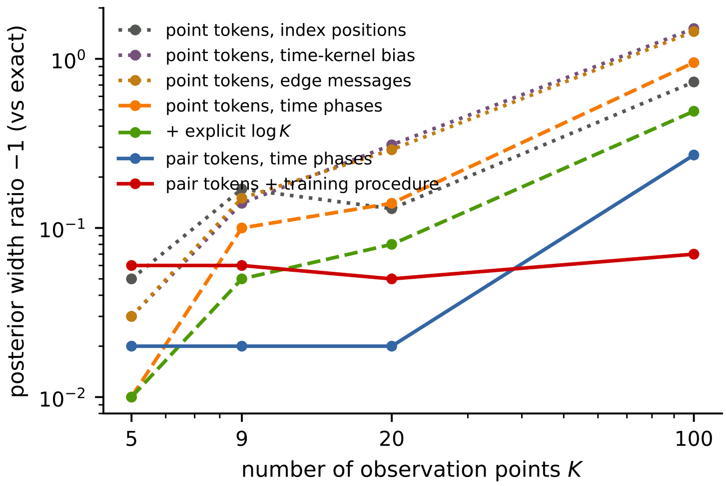

Figure 7: The data of Table 14 in graphical form: excess posterior width for the GBM volatility — the width ratio against the exact posterior minus one, so 0.1 means a posterior 10% too wide and lower is better — as a function of the number of observation points K (both axes logarithmic). Dotted curves: point-token variants with no set-size information, which degrade as designs become dense; dashed: partial repairs; solid: increment tokens, and increment tokens combined with the training procedure of Sec. 5, which is flat in K.